

# Faster Even-Cycle Listing via Unbalanced Supersaturation

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## ABSTRACT

This thesis studies the fundamental problem of listing even-length cycles through the lens of extremal graph theory. A central theme is that structural information on the number of even-length cycles in bipartite graphs can be converted into running-time bounds for cycle-listing algorithms.

The main result is a weak form of the Unbalanced Supersaturation Conjecture for even cycles. We prove that a bipartite graph  $G = (A \cup B, E)$  with sufficiently many edges relative to the threshold  $\max\{|A|, |B|\} \cdot \min\{|A|, |B|\}^{1/k}$  must contain many copies of  $C_{2k}$ . Although this result constitutes a step toward the full conjecture, it already provides sufficient information for the algorithmic applications considered in this work.

Using this extremal bound, we revisit the analysis framework of Williams and Westover for listing even cycles. This yields a refined running-time analysis that matches the state of the art for listing  $C_6$ , while avoiding the computer-assisted analysis used in their work [23]. Moreover, it improves the previously known state-of-the-art bound for listing  $C_{2k}$  for every  $k \geq 4$ , due to Alon, Yuster, and Zwick [23, 4].

We also extend the discussion to hypergraphs. The incidence graph gives a global reduction from listing Berge cycles of fixed length  $\ell$  to listing graph cycles of length  $2\ell$ . In addition, a direct labelled-2-shadow reduction lists all Berge cycles of length  $\ell$  passing through a prescribed hypergraph vertex.



Η παρούσα διπλωματική εργασία μελετά το θεμελιώδες πρόβλημα της απαρίθμησης κύκλων άρτιου μήκους υπό το πρίσμα της ακραίας θεωρίας γραφημάτων. Κεντρική ιδέα είναι ότι η δομική πληροφορία για τον αριθμό των κύκλων άρτιου μήκους σε διμερή γραφήματα μπορεί να μετατραπεί σε φράγματα για τον χρόνο εκτέλεσης αλγορίθμων απαρίθμησης κύκλων.

Κύριο αποτέλεσμα της εργασίας είναι μια ασθενής μορφή της Εικασίας Μη Ισορροπημένου Υπερκορεσμού για άρτιους κύκλους. Αποδεικνύουμε ότι ένα διμερές γράφημα  $G = (A \cup B, E)$  με επαρκώς πολλές ακμές ως προς το κατώφλι  $\max\{|A|, |B|\} \cdot \min\{|A|, |B|\}^{1/k}$  πρέπει να περιέχει πολλές εμφανίσεις  $C_{2k}$ . Παρότι το αποτέλεσμα αυτό αποτελεί ένα βήμα προς την πλήρη εικασία, παρέχει ήδη επαρκή πληροφορία για τις αλγοριθμικές εφαρμογές που εξετάζονται στην παρούσα εργασία.

Χρησιμοποιώντας αυτό το ακραίο φράγμα, επανεξετάζουμε το πλαίσιο ανάλυσης των Williams και Westover για την απαρίθμηση άρτιων κύκλων. Αυτό οδηγεί σε μια βελτιωμένη ανάλυση του χρόνου εκτέλεσης, η οποία ισοφαρίζει την τρέχουσα βέλτιστη επίδοση για την απαρίθμηση  $C_6$ , αποφεύγοντας όμως τη χρήση υπολογιστή [23]. Επιπλέον, βελτιώνει το προηγούμενως γνωστό βέλτιστο φράγμα για την απαρίθμηση  $C_{2k}$  για κάθε  $k \geq 4$ , το οποίο οφείλεται στους Alon, Yuster και Zwick [23, 4].

Επεκτείνουμε επίσης τη συζήτηση στα υπεργραφήματα. Το γράφημα πρόσπτωσης δίνει μια γενική αναγωγή από την απαρίθμηση κύκλων Berge σταθερού μήκους  $\ell$  στην απαρίθμηση κύκλων μήκους  $2\ell$  σε απλά γραφήματα. Επιπλέον, δίνουμε μια άμεση αναγωγή μέσω του επισημασμένου 2-σκειώδες γραφήματος για την απαρίθμηση όλων των κύκλων Berge μήκους  $\ell$  που περνούν από μια προκαθορισμένη κορυφή του υπεργραφήματος.



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# CHAPTER 1

## INTRODUCTION

### 1.1 Problem Setting and Motivation

For a fixed integer  $k \geq 2$ , we study the problem of listing all copies of cycles with length  $2k$  in a graph  $G$ . More precisely, given a graph  $G$  with  $m$  edges, the goal is to output all copies of  $C_{2k}$  contained in  $G$ , and to do so as efficiently as possible as a function of  $m$  and the output size. Since the number of copies may itself be large, the natural objective is an output-sensitive running time of the form

$$\text{preprocessing cost} + t,$$

where  $t$  denotes the number of copies of  $C_{2k}$  in  $G$  (not necessarily known).

To place the listing problem in context, we begin by considering the broader complexity landscape of detecting cycles of varying length. The algorithmic complexity of finding a cycle of length  $\ell$  varies significantly with  $\ell$ . At one extreme, when  $\ell = n$ , the problem becomes the Hamiltonian cycle problem, one of Karp's original 21 NP-complete problems [15]. More generally, deciding whether a graph contains a cycle of length  $\ell$  remains NP-complete when  $\ell = n^{\Omega(1)}$  [10, 8]. At the other extreme, when  $\ell = O(1)$ , the problem is fixed-parameter tractable<sup>1</sup>: Monien gave an  $O(f(\ell)m)$ -time algorithm for deciding whether a given vertex is contained in a  $C_\ell$  [20]. For odd constant  $\ell$ , the fastest known algorithms are based on matrix multiplication [4]. For even constant  $\ell$ , however, sharper bounds are possible [25, 8]. The present thesis focuses on this latter regime in the setting of output-sensitive listing.

Cycle and subgraph listing problems are natural from an applications perspective. Small cycles are useful in the analysis of complex networks [19] and in the study of social networks [11]. They also appear in computational biology and in biological interaction graphs [16]. Additionally, fixed subgraph patterns play an important role in graph pattern querying and related data-management tasks [9].

In these settings, the task is often inherently enumerative, as one is interested in identifying all occurrences of a fixed pattern. This makes output-sensitive listing algo-

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<sup>1</sup>Informally, a problem is fixed-parameter tractable (FPT) with respect to a parameter  $k$  if it can be solved in time  $f(k) \cdot n^{O(1)}$ , where  $f$  may depend arbitrarily on  $k$ , but the dependence on the input size  $n$  remains polynomial.

rithms particularly natural, since their running time is allowed to depend on the number  $t$  of reported patterns, while still aiming to keep the non-output portion of the computation as small as possible.

Even cycles occupy a distinguished position in both graph theory and graph algorithms. On the structural side, they are among the most classical objects in extremal graph theory, with deep results relating edge density to the existence and multiplicity of copies of  $C_{2k}$  [6, 12]. On the algorithmic side, they form one of the basic fixed-pattern listing problems, and their complexity is governed by the interplay between sparsity and the length of the cycle [3, 4, 8, 23].

From now on our focus is on sparse graphs. This is not a limitation of the present approach, but rather a consequence of the form of the running-time bound itself. In dense graphs, the non-output part of the running time is eventually dominated by the output term  $t$ , since sufficiently many edges already force many copies of  $C_{2k}$ . Thus, the sparse regime is the natural setting in which improved structural information can lead to genuinely faster listing algorithms. This intuition will be made precise in Chapter 3, where we discuss the sparse-dense threshold and explain the role of the Unbalanced Supersaturation Conjecture in the conditional optimality of the  $C_{2k}$ -listing problem.

## 1.2 Background and Prior Work

This section reviews the main prior work relevant to the thesis. Since our results lie at the interface of extremal graph theory and graph algorithms, we organize the discussion accordingly. We first survey the extremal results that motivate the supersaturation questions studied later, and then turn to the algorithmic results on detecting and listing even cycles.

### 1.2.1 Extremal graph theory background

For a fixed integer  $k \geq 2$ , one of the central questions in extremal graph theory is the determination of the Turán number  $\text{ex}(n, C_{2k})$ , that is, the maximum number of edges in an  $n$ -vertex graph containing no copy of  $C_{2k}$ . The classical upper bound is due to Bondy and Simonovits, who proved that every  $n$ -vertex graph with at least  $100k n^{1+1/k}$  edges contains a cycle  $C_{2k}$  [6]. In particular,  $\text{ex}(n, C_{2k}) = O(n^{1+1/k})$ . This theorem is one of the cornerstones of extremal graph theory.

A long line of work has sought to sharpen the constant and improve the general upper bound for  $\text{ex}(n, C_{2k})$ . For the purposes of this thesis, it is enough to note that the exponent  $1 + 1/k$  is the correct one, while the precise constant remains a subtle question. A recent general improvement is due to He, who obtained the bound

$$\text{ex}(n, C_{2k}) \leq (16\sqrt{5}\sqrt{k \log k} + o(1))n^{1+1/k}$$

for fixed  $k$  [12]. More generally, the literature on the Turán number of even cycles is extensive, and surveys such as [22] place these developments in a broader extremal context.

From the perspective of this thesis, the Bondy-Simonovits theorem is important not only as a Turán-number statement, but also as an *existential threshold* for even cycles in general graphs. Once the number of edges exceeds  $n^{1+1/k}$  by a sufficiently large constant factor, the graph must contain a copy of  $C_{2k}$ .

A stronger question is the corresponding *counting* problem: once the graph lies above the extremal threshold, how many copies of  $C_{2k}$  must it contain? In this direction, Simonovits proved a supersaturation theorem asserting that for every fixed  $k$ , graphs with sufficiently many edges above the Bondy-Simonovits threshold contain many copies of  $C_{2k}$ . This means that there exist constants  $C, c > 0$ , depending only on  $k$ , such that every  $n$ -vertex graph  $G$  with  $m \geq Cn^{1+1/k}$  contains at least  $c \left(\frac{m}{n}\right)^{2k}$  copies of  $C_{2k}$  [13]. Jiang and Yepremyan prove a hypergraph generalization of this statement for even linear cycles, and their argument includes the graph case as  $r = 2$ .

The bipartite setting is more directly connected to the algorithmic side of the thesis. Dahlgaard, Knudsen, and Stöckel proved a bipartite analogue of the Bondy-Simonovits existence theorem. Their result states that if a bipartite graph with sides of sizes  $L$  and  $R$  has more than

$$100k(L + R + (LR)^{(k+1)/(2k)})$$

edges, then it must contain a copy of  $C_{2k}$  [8, 23]. Thus, the *existence* problem for even cycles is already well understood in the bipartite setting.

What remains open is the corresponding *counting* statement in bipartite graphs. This is precisely the role of the Unbalanced Supersaturation Conjecture. As formulated by Williams and Westover, following a communication of Jin and Zhou, the conjecture asserts that if a bipartite graph  $G = (A \cup B, E)$  satisfies

$$|E| \geq 100k(|A| + |B| + (|A||B|)^{(k+1)/(2k)}),$$

then the number of copies of  $C_{2k}$  in  $G$  should satisfy

$$\#C_{2k}(G) \geq \Omega\left(\frac{|E|^{2k}}{|A|^k |B|^k}\right)$$

for every fixed  $k$  [23]. This is the missing bipartite counting theorem that would complete the extremal picture relevant to sparse even-cycle listing.

The importance of this conjecture is not purely extremal. Williams and Westover prove that such an unbalanced supersaturation statement is exactly the input needed to obtain optimal running-time bounds in their layered graph analysis framework [23] for listing even length cycles.

The special cases  $k = 2$  and  $k = 3$  are particularly informative. For  $k = 2$ , the problem is related to the listing of  $C_4$  and is essentially solved [2, 23]. For  $k = 3$ , Williams and Westover prove a partial supersaturation theorem for  $C_6$ 's in bipartite graphs, strong enough to support their sparse 6-cycle listing result [23].

In summary, as shown in Table 1.1, the theory of even cycles in simple graphs is classical at both the existential and the counting level. In bipartite graphs, the existence theory is already in place, but the corresponding counting theory remains incomplete. This gap is exactly where the present thesis is situated.

### 1.2.2 Algorithmic background

We now turn to the algorithmic side. Here the relevant distinction is between *detection* and *listing*. Detection asks whether the input graph contains a copy of  $C_{2k}$ , while listing asks that all copies be output. The latter is naturally more demanding, and the strongest bounds are typically output-sensitive.

For fixed  $k$ , even-cycle detection is well understood in terms of the number of vertices. Yuster and Zwick showed that one can detect a cycle  $C_{2k}$  in  $O(n^2)$  time for

| Setting          | Existence                                                                                                                  | Counting / supersaturation                                                                                                                                           |
|------------------|----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| General graphs   | Bondy-Simonovits: if $m \geq C(k)n^{1+1/k}$ , then $G$ contains a copy of $C_{2k}$ [6].                                    | Simonovits; Jiang-Yepremyan: if $m \geq C(k)n^{1+1/k}$ , then $\#C_{2k}(G) \geq c(k)(m/n)^{2k}$ [13].                                                                |
| Bipartite graphs | Dahlgaard-Knudsen-Stöckel: if $m \geq C(k)( A  +  B  + ( A  B )^{(k+1)/(2k}))$ , then $G$ contains a copy of $C_{2k}$ [8]. | Unbalanced Supersaturation Conjecture: if $m \geq C(k)( A  +  B  + ( A  B )^{(k+1)/(2k}))$ , then $\#C_{2k}(G) \geq c(k)m^{2k}/( A ^k B ^k)$ ; open in general [23]. |

Table 1.1: Existence and counting results for even cycles in general and bipartite graphs. Here  $n$  denotes the number of vertices of the graph,  $m$  the number of edges, and  $A, B$  the two sides of the bipartition in the bipartite setting.

every fixed  $k \geq 2$  [25]. This  $n$ -based bound serves as the dense-graph benchmark for the problem.

In terms of the number of edges  $m$ , earlier sparse-graph bounds were obtained by Alon, Yuster, and Zwick in their work on finding and counting fixed-length cycles [4]. Dahlgaard, Knudsen, and Stöckel then made the key improvement for even-cycle detection: they gave an algorithm running in  $O(m^{2k/(k+1)})$  time for detecting a  $C_{2k}$  in a graph with  $m$  edges [8]. They also showed that this bound is conditionally optimal under the hypothesis that the  $O(n^2)$  detection bound is optimal in terms of  $n$ . Thus, even-cycle detection may be regarded as essentially solved, both in terms of  $n$  and in terms of  $m$ .

The main conceptual contribution of Dahlgaard, Knudsen, and Stöckel is the introduction of *capped  $k$ -walks*. Their running-time analysis is governed by the number of capped  $k$ -walks, that is, walks of length  $k$  that respect a fixed ordering on the vertices. This viewpoint is fundamental for the later listing framework of Williams and Westover [8, 23].

On the listing side, the picture is more recent and more delicate. The case  $k = 2$ , that is, listing  $C_4$ 's, is the only one that is currently understood essentially optimally. Abboud, Khoury, Leibowitz, and Safier proved that all 4-cycles in a graph can be listed in time  $\tilde{O}(\min\{n^2, m^{4/3}\} + t)$ , and showed that this is best possible up to polylogarithmic factors under standard fine-grained assumptions [2]. Thus,  $C_4$ -listing is the model case in which the sparse output-sensitive complexity is already sharply understood.

For  $k = 3$ , the problem of listing  $C_6$ 's has seen rapid recent progress. Jin, Vasilevska Williams, and Zhou gave an  $\tilde{O}(n^2 + t)$ -time algorithm for listing all 6-cycles in an  $n$ -vertex graph [14]. This nearly matches the quadratic detection bound of Yuster and Zwick in the  $n$ -vertex model. Williams and Westover then considered the sparse setting and proved that  $C_6$ 's can be listed in time  $\tilde{O}(m^{1.6} + t)$ , improving the previous  $O(m^{5/3} + t)$  bound from the older Alon-Yuster-Zwick line [23]. Their proof is highly nontrivial: it relies on a partial supersaturation theorem for hexagons and a computer-assisted case analysis over hundreds of linear programs.

For larger even cycles, the sparse listing problem is much less complete. Williams and Westover state that the previous sparse state of the art for listing  $C_{2k}$ 's, for  $k \geq 4$ , is the bound  $O(m^{2-1/k} + t)$ , which follows from the work of Alon, Yuster, and Zwick [23, 4]. They further explain that the Unbalanced Supersaturation Conjecture would imply the conjecturally optimal bound  $\tilde{O}(m^{2k/(k+1)} + t)$  for all fixed  $k$ . In other

words, once the correct unbalanced counting theorem is available, the layered graph framework is already strong enough to yield the desired sparse listing algorithm.

For odd cycles, the situation is rather different. Unlike even cycles, odd cycles do not admit a comparable sparsity threshold: a bipartite graph may have  $\Theta(n^2)$  edges and still contain no odd cycle. Consequently, the fastest known algorithms for detecting fixed odd cycles are matrix-multiplication based. Triangle detection can be solved in  $O(n^\omega)$  time, where  $\omega$  denotes the matrix multiplication exponent, and more generally Alon, Yuster, and Zwick showed that for every constant  $\ell$ , a cycle  $C_\ell$  can be found in  $O(n^\omega)$  expected time and  $O(n^\omega \log n)$  deterministic time [3, 4]. As noted in the discussion of Dahlgaard, Knudsen, and Stöckel, when  $\ell$  is odd, this remains the fastest known general bound [8]. This stands in sharp contrast to the even-cycle setting, where extremal results imply that sufficiently dense graphs must already contain a  $C_{2k}$ , making substantially faster sparse-graph algorithms possible.

In fine-grained complexity, it is standard to assume widely believed conjectures and use reductions to argue that substantially faster algorithms for basic problems are unlikely. This perspective already appears in the cycle problems discussed above. For example, Dahlgaard, Knudsen, and Stöckel note that their  $O(m^{2k/(k+1)})$ -time algorithm for detecting  $C_{2k}$  is conditionally optimal for  $k \equiv 2 \pmod{4}$  among algorithms that do not use fast matrix multiplication, citing reductions of Lincoln and Vyas [8, 18]. In particular, Williams and Westover recall that, for sufficiently large constant  $k$ , an  $m^{3/2-\varepsilon}$ -time algorithm for detecting  $C_{2k}$  in sparse graphs would imply a  $2^{(1-\varepsilon')n} \text{poly}(n)$ -time algorithm for Max-3-SAT, which is considered unlikely [23, 18]. Conditional hardness is also known on the listing side: the nearly optimal bound  $\tilde{O}(\min\{n^2, m^{4/3}\} + t)$  for listing  $C_4$ 's was shown to be conditionally optimal under the 3SUM hypothesis, and later even under the weaker Exact Triangle hypothesis [2, 14, 1, 7]. More broadly, the literature on triangle listing already exhibits the same pattern: upper bounds are matched, in appropriate regimes, by lower bounds under 3SUM and related hypotheses [14, 21, 17, 24].

To summarize, as shown in Table 1.2, the problem of detecting even cycles admits optimal algorithms both in terms of  $n$  and in terms of  $m$ . For the listing problem, obtaining an optimal algorithm in terms of  $m$  would follow from a solution to the Unbalanced Supersaturation Conjecture.

### 1.3 Our Contribution

The contributions of this thesis are threefold.

First, we prove new extremal results for even cycles in bipartite graphs. The main extremal contribution is a weak form of the Unbalanced Supersaturation Conjecture: if a bipartite graph  $G = (A \cup B, E)$  has sufficiently many edges relative to the threshold  $\max\{|A|, |B|\} \min\{|A|, |B|\}^{1/k}$ , then it contains many copies of  $C_{2k}$ . This does not establish the full conjecture, but it gives a supersaturation bound strong enough for the algorithmic applications considered here. In addition, we give an elementary graph-theoretic proof of the folklore case  $k = 2$ , thereby proving the Unbalanced Supersaturation Conjecture for  $C_4$ 's.

Second, we use these extremal results to refine the analysis of even-cycle listing algorithms. Using the layered graph framework of Williams and Westover [23], we

| Problem                                               | Best known bound                      | Reference / comment                                                              |
|-------------------------------------------------------|---------------------------------------|----------------------------------------------------------------------------------|
| Detection of $C_{2k}$ in terms of $n$                 | $O(n^2)$                              | Yuster-Zwick [25].                                                               |
| Detection of $C_{2k}$ in terms of $m$                 | $O(m^{2k/(k+1)})$                     | Dahlgaard-Knudsen-Stöckel; conditionally optimal [8].                            |
| Detection of $C_{2k+1}$ for fixed $k$ (deterministic) | $O(n^\omega \log n)$                  | Alon-Yuster-Zwick; obtained via color-coding and fast matrix multiplication [4]. |
| Listing $C_4$                                         | $\tilde{O}(\min\{n^2, m^{4/3}\} + t)$ | Abboud-Khoury-Leibowitz-Safier; essentially optimal [2].                         |
| Listing $C_6$ in terms of $n$                         | $\tilde{O}(n^2 + t)$                  | Jin-Vassilevska Williams-Zhou [14].                                              |
| Listing $C_6$ in terms of $m$                         | $\tilde{O}(m^{1.6} + t)$              | Williams-Westover [23].                                                          |
| Listing $C_{2k}$ , $k \geq 4$                         | $O(m^{2-1/k} + t)$                    | Previous sparse state of the art, via Alon-Yuster-Zwick [23, 4].                 |

Table 1.2: Selected algorithmic results for detecting and listing fixed-length cycles. The even-cycle results are the main focus of this thesis; the row for odd cycles is included for comparison.

obtain improved bounds on the number of capped walks and hence improved output-sensitive listing bounds. For  $k = 2$ , the conditionally optimal  $C_4$ -listing algorithm follows directly from the  $k = 2$  solution of the Unbalanced Supersaturation Conjecture, placing the known  $C_4$ -listing bound inside the same unified framework. For  $k = 3$ , our analysis recovers the current best bound for listing  $C_6$ 's without the computer-aided part of the previous analysis. For every  $k \geq 4$ , it improves the previously known sparse  $m$ -based bound for listing  $C_{2k}$ 's.

Third, we extend the viewpoint to hypergraphs. The incidence graph gives a global reduction from listing Berge cycles of fixed length  $\ell$  to listing graph cycles of length  $2\ell$ . We also give a direct labelled-2-shadow reduction for the rooted version of the problem, listing all Berge cycles of length  $\ell$  that pass through a prescribed hypergraph vertex without doubling the cycle length.

## 1.4 Technical Overview

The technical development of the thesis is carried out in Chapters 3, 4, and 5.

Chapter 3 presents the algorithmic framework. We start from the color-coding method of Alon, Yuster, and Zwick [3], the capped-walk framework of Dahlgaard, Knudsen, and Stöckel [8], and the adaptation of this framework from detection to listing by Williams and Westover [23]. This leads to a layered graph formulation in which the running-time analysis reduces, up to polylogarithmic factors, to bounding the number of capped walks in an auxiliary layered graph. We then introduce the Unbalanced Supersaturation Conjecture and explain why such extremal estimates are the input needed to control this quantity.

Chapter 4 supplies the extremal input. We first prove the  $k = 2$  case of the Unbalanced Supersaturation Conjecture using an elementary counting argument for  $C_4$ 's. We then prove the weak form of the conjecture for general  $k$ , which is the main extremal theorem used later in the thesis. Although this theorem is weaker than the full conjec-

ture, it is strong enough for the layered graph analysis. For completeness, an alternative proof of the main extremal theorem is included in the appendix.

Chapter 5 incorporates the extremal bounds into the Williams-Westover framework. The  $k = 2$  case gives a unified derivation of the conditionally optimal  $\tilde{O}(\min\{m^{4/3}, n^2\} + t)$  bound for  $C_4$ -listing. For  $k = 3$ , the same framework yields the current best  $\tilde{O}(m^{1.6} + t)$  bound for  $C_6$ -listing by an analytic argument. For  $k \geq 4$ , it gives improved  $m$ -based bounds for listing  $C_{2k}$ 's. The chapter concludes with the two hypergraph extensions: the incidence graph reduction and the labelled-2-shadow reduction for rooted Berge-cycle listing.



## CHAPTER 2

## PRELIMINARIES

In this chapter we collect the basic definitions and notation used throughout the thesis. Unless stated otherwise, all graphs are finite, simple, and undirected.

### 2.1 Graphs, Hypergraphs, and Cycles

**Definition 2.1** (Graph). A graph is a pair  $G = (V, E)$ , where  $V$  is a finite set of vertices and  $E \subseteq \binom{V}{2}$  is a set of edges. We write  $V(G)$  and  $E(G)$  for the vertex set and the edge set of  $G$ , respectively.

**Definition 2.2** (Bipartite graph). A graph  $G = (V, E)$  is bipartite if its vertex set can be written as  $V = A \cup B$ , where  $A \cap B = \emptyset$ , and every edge of  $G$  has one endpoint in  $A$  and one endpoint in  $B$ . In this case we write  $G = (A \cup B, E)$ .

**Definition 2.3** (Walk and path). A walk of length  $\ell$  in a graph  $G$  is a sequence of vertices  $v_0, v_1, \dots, v_\ell$  such that  $\{v_{i-1}, v_i\} \in E(G)$  for every  $i \in \{1, \dots, \ell\}$ . A path is a walk in which all vertices are distinct.

**Definition 2.4** (Cycle). A cycle of length  $\ell$  in a graph  $G$  is a sequence of distinct vertices  $v_0, v_1, \dots, v_{\ell-1}$  such that  $\{v_i, v_{i+1}\} \in E(G)$  for every  $i \in \{0, \dots, \ell-2\}$ , and also  $\{v_{\ell-1}, v_0\} \in E(G)$ . Thus, a cycle of length  $\ell$  has exactly  $\ell$  edges, and is denoted by  $C_\ell$ . In particular, an even cycle of length  $2k$  is denoted by  $C_{2k}$ .

**Definition 2.5** (Copy of  $C_{2k}$ ). A copy of  $C_{2k}$  in a graph  $G$  is a subgraph of  $G$  isomorphic to  $C_{2k}$ . We write  $\#C_{2k}(G)$  for the number of copies of  $C_{2k}$  contained in  $G$ .

**Definition 2.6** (Hypergraph). A hypergraph is a pair  $\mathcal{H} = (V, \mathcal{E})$ , where  $V$  is a finite set of vertices and  $\mathcal{E} \subseteq 2^V$  is a family of subsets of  $V$ , called hyperedges. The rank of  $\mathcal{H}$  is  $\max_{e \in \mathcal{E}} |e|$ .

**Definition 2.7** (Incidence graph). Let  $\mathcal{H} = (V, \mathcal{E})$  be a hypergraph. Its incidence graph is the bipartite graph  $I(\mathcal{H})$  with bipartition  $V \cup \mathcal{E}$ , in which a vertex  $v \in V$  is adjacent to a hyperedge  $e \in \mathcal{E}$  if and only if  $v \in e$ .

**Definition 2.8** (Berge cycle). A Berge cycle of length  $\ell$  in a hypergraph  $\mathcal{H} = (V, \mathcal{E})$  is a sequence  $v_1, e_1, v_2, e_2, \dots, v_\ell, e_\ell$  of distinct vertices  $v_1, \dots, v_\ell \in V$  and distinct

hyperedges  $e_1, \dots, e_\ell \in \mathcal{E}$  such that  $\{v_i, v_{i+1}\} \subseteq e_i$  for every  $i \in \{1, \dots, \ell\}$ , where the indices are taken modulo  $\ell$ .

## 2.2 Notation

We use standard asymptotic notation. Given functions  $f, g : \mathbb{N} \rightarrow \mathbb{R}_{\geq 0}$ , we write  $f = O(g)$  if there exists a constant  $C > 0$  such that  $f(n) \leq Cg(n)$  for all sufficiently large  $n$ . The notation  $\Omega(g)$  and  $\Theta(g)$  is defined as usual.

We write  $f = \tilde{O}(g)$  if there exists a constant  $C > 0$  such that  $f(n) = O(g(n)(\log n)^C)$ , that is,  $\tilde{O}(\cdot)$  suppresses polylogarithmic factors.

We write  $f \gtrsim g$  if there exists a constant  $C > 0$ , independent of the asymptotic parameters, such that  $f(n) \geq Cg(n)$  for all sufficiently large  $n$ ; equivalently,  $g = O(f)$ . When  $k$  is fixed, the implicit constants may depend on  $k$ .

## 2.3 Background Extremal Results

The following theorem is the supersaturation counterpart of the Bondy-Simonovits extremal theorem for even cycles [6]. In the graph case, this supersaturation statement goes back to Simonovits; Jiang and Yepremyan later gave a self-contained proof as the  $r = 2$  case of their more general theorem for linear hypergraphs [13].

**Theorem 2.9** (Simonovits; Jiang-Yepremyan). For every integer  $k \geq 2$ , there exist positive constants  $C, c > 0$ , depending only on  $k$ , such that the following holds. Let  $G = (V, E)$  be a graph with  $n := |V|$  and  $m := |E|$ . If

$$m \geq Cn^{1+1/k},$$

then  $G$  contains at least

$$c \left( \frac{m}{n} \right)^{2k}$$

copies of  $C_{2k}$ .

We will also use the following two-set edge bound of Dahlgaard, Knudsen, and Stöckel [8, Lemma 6]. It is useful because it immediately yields existential results in both settings: by taking  $A = B = V(G)$ , one obtains an existence threshold for general graphs, while by taking  $A$  and  $B$  to be the two sides of a bipartition, one obtains the corresponding bipartite existence result.

**Theorem 2.10** (Dahlgaard-Knudsen-Stöckel, Lemma 6). Let  $G = (V, E)$  be a graph, let  $A, B \subseteq V$ , and let  $k \geq 2$  be an integer. If  $G$  contains no copy of  $C_{2k}$ , then

$$|E(A, B)| \leq 100k \left( (|A||B|)^{\frac{k+1}{2k}} + |A| + |B| \right).$$

# CHAPTER 3

## ALGORITHMIC FRAMEWORK AND PREVIOUS RESULTS

In this chapter, we present the algorithmic framework on which the results of the thesis build. We first describe the underlying even-cycle listing algorithm and the earlier ideas from which it arises. We then introduce the layered graph framework used to analyze its running time, together with the Unbalanced Supersaturation Conjecture and its role in this analysis. Finally, we discuss the special cases  $k = 2$  and  $k = 3$ , which help illustrate the current landscape of the problem.

### 3.1 The Algorithm

We begin with the detection setting, since the listing algorithm considered later is a natural modification of it.

**Definition 3.1** (Degree ordering). Let  $G = (V, E)$  be a graph. A *degree ordering* of  $V$  is a total ordering  $\succ$  of the vertices such that

$$u \succ v \quad \text{whenever} \quad \deg(u) > \deg(v),$$

with ties broken arbitrarily.

**Definition 3.2** (Capped  $k$ -walk). Let  $\succ$  be a degree ordering of the vertices of a graph  $G$ . A  $\succ$ -*capped  $k$ -walk* is a walk

$$x_0, x_1, \dots, x_k \quad \text{such that} \quad x_0 \succ x_i \quad \text{for every } i \in [k].$$

When the ordering is clear from the context, we simply refer to such a walk as a *capped  $k$ -walk*.

Dahlgaard, Knudsen, and Stöckel [8] consider the following vertex-based detection problem: given a graph  $H$  and a vertex  $v \in V(H)$ , decide whether  $v$  belongs to a copy of  $C_{2k}$  in  $H$ . They assume access to an algorithm  $A$  for this task and use it as a black box inside their reduction.

Such a routine can be obtained in several ways. One example is Monien's algorithm [20], which gives a deterministic fixed-parameter algorithm for finding fixed-length paths and cycles, with running time  $O(k! \cdot |E(H)|)$ . Another example is the color-coding method of Alon, Yuster, and Zwick [3], which gives a randomized fixed-parameter algorithm with running time  $O(2^{O(k)} \cdot |E(H)|)$ , and can also be derandomized with only a small loss in efficiency.

Let

$$v_1 \prec v_2 \prec \dots \prec v_n$$

be the degree ordering of the vertices of  $G$ . For each  $i \in [n]$ , let  $G_{\leq i}$  denote the subgraph of  $G$  induced by  $\{v_1, \dots, v_i\}$ . Dahlgaard, Knudsen, and Stöckel then define  $G_i^{\leq k}$  to be the subgraph of  $G_{\leq i}$  induced by the vertices whose distance from  $v_i$  in  $G_{\leq i}$  is at most  $k$ .

Their reduction proceeds by considering the sequence of graphs

$$G_1^{\leq k}, G_2^{\leq k}, \dots, G_n^{\leq k},$$

and running the routine  $A$  on each pair  $(G_i^{\leq k}, v_i)$ . In this way, the global problem of detecting a copy of  $C_{2k}$  in  $G$  is reduced to a sequence of local vertex-based detection problems. The reduction of Dahlgaard, Knudsen, and Stöckel admits a natural modification in the listing setting. We record here the algorithm of Williams and Westover [23] in the form in which it will be used later.

**Lemma 3.3** (Listing colorful  $k$ -cycles through a fixed vertex). Let  $k \in \mathbb{N}$  be fixed, let  $G = (V, E)$  be a simple graph, let  $v \in V$ , and let  $c : V \rightarrow [k]$  be a vertex-coloring. Then there exists an algorithm that outputs all colorful cycles of length  $k$  passing through  $v$  in time  $O(|E| + t)$ , where  $t$  is the number of reported cycles.

*Proof.* Let  $a := c(v)$ , and consider all  $(k-1)!$  permutations  $\pi = (\pi_2, \dots, \pi_k)$  of the colors in  $[k] \setminus \{a\}$ .

For each permutation  $\pi$ , construct a layered graph  $H_\pi$  as follows. The first layer is  $L_1 = \{v\}$ . For  $2 \leq i \leq k-1$ , let  $L_i = \{u \in V : c(u) = \pi_i\}$ . The last layer is  $L_k = \{u \in N_G(v) : c(u) = \pi_k\}$ . We place an edge between  $x \in L_i$  and  $y \in L_{i+1}$  whenever  $\{xy\} \in E(G)$ .

Every path  $v = x_1, \dots, x_k$  in  $H_\pi$  gives a colorful cycle  $v = x_1, \dots, x_k, v$  in  $G$ . Conversely, every colorful  $k$ -cycle through  $v$  arises in this way for some permutation  $\pi$ . Since the layers correspond to distinct colors, all vertices on the path are distinct.

Thus, for each permutation  $\pi$ , it suffices to enumerate all paths from  $L_1$  to  $L_k$  in  $H_\pi$ . This takes time  $O(|E| + t_\pi)$ , where  $t_\pi$  is the number of cycles reported for  $\pi$ . To avoid duplicates, we either deduplicate explicitly or keep only one permutation from each reversal pair. Since  $k$  is fixed, the number of permutations is constant, and so the total running time is

$$O((k-1)! (|E| + t_1 + \dots + t_{(k-1)!})) = O(|E| + t),$$

where  $t = \sum_\pi t_\pi$  is the total number of reported cycles.  $\square$

**Theorem 3.4.** Fix a parameter  $\Delta \in \mathbb{N}$ , and let  $G$  be a graph. Let  $W$  be the number of capped  $k$ -walks in  $G$  that start at a vertex of degree at most  $\Delta$ . Then there is a randomized algorithm to list all copies of  $C_{2k}$  in  $G$  in time

$$\tilde{O}\left(\frac{m^2}{\Delta} + W + t\right),$$

where  $t$  is the number of copies of  $C_{2k}$  in  $G$ .

*Proof.* Label the vertices  $v_1, \dots, v_n$  so that  $v_i \succ v_j$  if and only if  $i > j$ . For each  $i \in [n]$ , let  $G_i$  denote the subgraph of  $G$  induced by  $\{v_1, \dots, v_i\}$ . We then define  $G_i^{\leq k}$  to be the subgraph of  $G_i$  whose vertices are the vertices at distance at most  $k$  from  $v_i$  in  $G_i$ , and whose edges are the edges between such vertices with at most one endpoint at distance exactly  $k$  from  $v_i$ . Equivalently,  $G_i^{\leq k}$  keeps the vertices and edges that are relevant for listing cycles of length  $2k$  containing  $v_i$ . For  $i = 1, 2, \dots, n$ , we list the copies of  $C_{2k}$  in  $G_i^{\leq k}$  that contain  $v_i$ .

To list these cycles, we use color-coding. Color each vertex independently and uniformly at random with a color from  $[2k]$ , and restrict attention to *colorful* copies of  $C_{2k}$ , that is, copies whose vertices receive pairwise distinct colors. For a fixed coloring, using the previous lemma 3.3, the colorful copies of  $C_{2k}$  in  $G_i^{\leq k}$  that contain  $v_i$  can be listed in time

$$O(|E(G_i^{\leq k})| + t_i).$$

Each copy of  $C_{2k}$  becomes colorful with probability  $\Omega(1)$ , since  $k$  is fixed. Hence, after repeating this procedure  $\tilde{O}(1)$  times, all copies are listed with high probability.

The correctness of the algorithm follows from the ordering of the vertices. Indeed, every copy of  $C_{2k}$  in  $G$  has a unique largest vertex with respect to  $\succ$ , say  $v_i$ . Since all vertices of the cycle lie in  $\{v_1, \dots, v_i\}$  and are at distance at most  $k$  from  $v_i$  along the cycle, the entire cycle is contained in  $G_i^{\leq k}$ . Therefore it is listed when the algorithm processes  $v_i$ , and since  $v_i$  is uniquely determined, each copy of  $C_{2k}$  is reported exactly once.

We now analyze the running time of the algorithm. Note that  $|E(G_i^{\leq k})|$  is at most the number of capped  $k$ -walks that start at  $v_i$ . Let  $T$  be the largest index such that  $\deg(v_T) \leq \Delta$ . Then, by definition,

$$\sum_{i=1}^T |E(G_i^{\leq k})| \leq O(W).$$

There are at most  $m/\Delta$  vertices of degree larger than  $\Delta$ . Thus,

$$\sum_{i=T+1}^n |E(G_i^{\leq k})| \leq O\left(\frac{m^2}{\Delta}\right).$$

Finally, we have

$$\sum_{i=1}^n t_i = O(t).$$

Combining these bounds, we conclude that the algorithm runs in time

$$\tilde{O}\left(t + W + \frac{m^2}{\Delta}\right).$$

□

## 3.2 The Framework for the Analysis

The goal of the analysis framework is to bound the number of capped  $k$ -walks in the graph. This is the key quantity governing the running time of the algorithm described in the previous section. The framework, introduced by Williams and Westover [23], organizes the graph into a suitable layered structure and reduces the running-time analysis to

bounding the number of capped walks that can appear in this structure. In this way, the algorithmic problem is translated into a combinatorial one, and extremal information on even cycles can be used to obtain improved running-time bounds.

**Lemma 3.5** (Layered decomposition, adapted from [23]). Let  $G$  be a graph. Then there exist a value  $d^*$ , a set  $V^* \subseteq V(G)$ , an induced subgraph  $G' \subseteq G$ , sets

$$X_1 \subseteq V^*, X_2, \dots, X_k \subseteq V(G'),$$

and integers

$$d_1, \dots, d_k \in \mathbb{N},$$

such that, writing  $X_{k+1} := V(G')$ , the following hold.

1. The number of capped  $k$ -walks in  $G$  is at most  $O(\log^{k+1} n)$  times the number of  $k$ -walks in

$$X_1 \times X_2 \times \dots \times X_k \times V(G').$$

2.  $V^*$  consists of vertices whose degrees in  $G$  lie in the interval  $[d^*/2, d^*]$ , and  $G'$  is the induced subgraph of  $G$  on the vertices of degree at most  $d^*$ .
3. For every  $i \in [k]$  and every  $v \in X_i$ , we have

$$|N_{X_{i+1}}(v)| \in [d_i/2, d_i].$$

*Proof.* Partition the vertices of  $G$  into dyadic degree classes. For each integer  $j$ , let  $V_j$  denote the set of vertices whose degree in  $G$  lies in the interval  $[2^{j-1}, 2^j]$ . Choose an index  $j^*$  for which the number of capped  $k$ -walks starting at a vertex of  $V_{j^*}$  is maximum. Define

$$V^* := V_{j^*}, \quad d^* := 2^{j^*},$$

and let  $G'$  be the induced subgraph of  $G$  on the vertices of degree at most  $d^*$ .

Every capped  $k$ -walk starts at some dyadic degree class, and by the choice of  $j^*$ , the number of capped  $k$ -walks in  $G$  is at most  $O(\log n)$  times the number of capped  $k$ -walks that start in  $V^*$ . Moreover, if a capped  $k$ -walk starts at a vertex of  $V^*$ , then all of its vertices lie in  $G'$ , since every later vertex in the walk has degree at most that of the first one. Hence the number of capped  $k$ -walks in  $G$  is at most  $O(\log n)$  times the number of  $k$ -walks in  $G'$  that start in  $V^*$ .

We now refine these walks by repeatedly grouping vertices according to dyadic ranges of their numbers of neighbors in the next layer. First, partition the vertices of  $G'$  according to their degree in  $G'$ : for each relevant integer  $i_k$ , let  $W_{i_k}^{(k)}$  be the set of vertices whose degree in  $G'$  lies in the  $i_k$ -th dyadic interval. Then, working backwards for  $\ell = k-1, k-2, \dots, 2$ , and for every choice of indices  $i_k, \dots, i_{\ell+1}$ , define  $W_{i_k, \dots, i_\ell}^{(\ell)}$  to be the set of vertices  $v \in V(G')$  for which the number of neighbors of  $v$  in  $W_{i_k, \dots, i_{\ell+1}}^{(\ell+1)}$  lies in the  $i_\ell$ -th dyadic interval. Finally, for each choice of  $i_k, \dots, i_2$ , define  $W_{i_k, \dots, i_1}^{(1)}$  analogously inside  $V^*$ , using the number of neighbors in  $W_{i_k, \dots, i_2}^{(2)}$ .

Every  $k$ -walk in  $G'$  starting in  $V^*$  belongs to one of the classes

$$W_{i_k, \dots, i_1}^{(1)} \times W_{i_k, \dots, i_2}^{(2)} \times \dots \times W_{i_k}^{(k)} \times V(G').$$

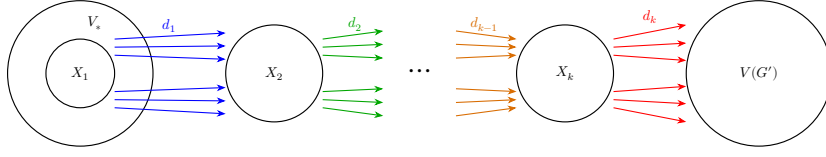


Figure 3.1: Layered decomposition from Lemma 3.5.

There are at most  $O(\log^k n)$  such classes. Choose indices  $(i_k, \dots, i_1)$  that maximize the number of  $k$ -walks in the corresponding class, and set

$$X_\ell := W_{i_k, \dots, i_\ell}^{(\ell)} \quad \text{for } \ell = 1, \dots, k,$$

and

$$d_\ell := 2^{i_\ell} \quad \text{for } \ell = 1, \dots, k.$$

By construction, every vertex  $v \in X_\ell$  has between  $d_\ell/2$  and  $d_\ell$  neighbors in  $X_{\ell+1}$ . This gives property (3). Property (2) holds by the definitions of  $V^*$  and  $G'$ .

Finally, because we selected the class maximizing the number of walks and there are only  $O(\log^k n)$  possible classes, the number of  $k$ -walks in

$$X_1 \times X_2 \times \dots \times X_k \times V(G')$$

is at least an  $\Omega(1/\log^k n)$ -fraction of the  $k$ -walks in  $G'$  starting in  $V^*$ . Combining this with the earlier  $O(\log n)$ -loss when restricting capped  $k$ -walks in  $G$  to walks in  $G'$  starting in  $V^*$ , we conclude that the number of capped  $k$ -walks in  $G$  is at most  $O(\log^{k+1} n)$  times the number of  $k$ -walks in

$$X_1 \times X_2 \times \dots \times X_k \times V(G'),$$

which proves property (1).  $\square$

Figure 3.1 illustrates the layered decomposition used in the proof.

### 3.3 The Unbalanced Supersaturation Conjecture and its Role

The Unbalanced Supersaturation Conjecture, abbreviated *USSC*, is the bipartite counting analogue of the existential theorem of Dahlgaard, Knudsen, and Stöckel [8]. In the form stated by Williams and Westover [23], it asserts that once a bipartite graph exceeds the natural extremal threshold, it must contain many copies of  $C_{2k}$ .

**Conjecture 3.6** (Unbalanced Supersaturation Conjecture). Let  $G = (A \cup B, E)$  be a bipartite graph, and let  $m := |E(G)|$ . Suppose that

$$m \geq 100k \left( |A| + |B| + (|A||B|)^{\frac{k+1}{2k}} \right).$$

Then

$$\#C_{2k}(G) \geq \Omega \left( \frac{m^{2k}}{|A|^k |B|^k} \right).$$

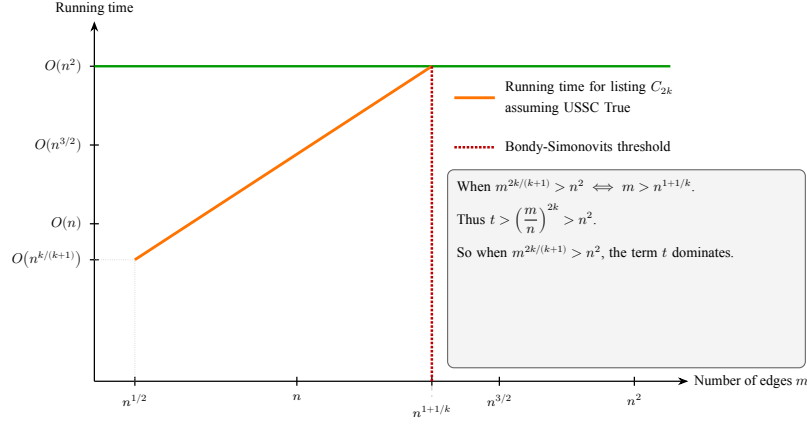


Figure 3.2: Comparison between the conjectured sparse running time for listing  $C_{2k}$  and the dense  $O(n^2)$  benchmark.

Its significance for this thesis is algorithmic. Williams and Westover show that, if one combines the listing algorithm from Section 3.1 with the capped-walk analysis framework of Section 3.2, then the conjecture yields a conditionally optimal algorithm for listing  $C_{2k}$ 's. More precisely, assuming Conjecture 3.6, one obtains a running time of

$$\tilde{O}\left(m^{\frac{2k}{k+1}+t}\right)$$

for listing all copies of  $C_{2k}$  [23]. Thus, the conjecture provides exactly the extremal input needed to obtain the conjectured optimal sparse bound within this framework.

To make the distinction between the sparse and dense regimes explicit, we take  $n^{1+1/k}$  as the natural edge threshold for the problem of listing  $C_{2k}$ 's (up to constant factors), in view of the Bondy-Simonovits existential threshold. Our interest in the sparse regime stems from the fact that beyond this threshold, the output term already dominates the running time. To make that more precise, if  $m^{\frac{2k}{k+1}} \gtrsim n^2$ , or equivalently  $m \gtrsim n^{1+1/k}$ , then the supersaturation bound gives  $t \gtrsim c(k) \left(\frac{m}{n}\right)^{2k} \gtrsim n^2$ .

In other words, once  $m^{\frac{2k}{k+1}}$  exceeds  $n^2$ , the number of copies of  $C_{2k}$  is already larger than the dense-graph benchmark. Therefore, algorithms with preprocessing term  $m^{\frac{2k}{k+1}}$  are most meaningful in the sparse regime, where improvements in the non-output part of the running time can lead to genuinely faster listing algorithms.

Figure 3.2 illustrates the relation between the conjectured sparse running time and the dense  $O(n^2)$  benchmark.

### 3.4 Special Cases: $k = 2$ and $k = 3$

The cases  $k = 2$  and  $k = 3$  are particularly informative, since they illustrate the current state of the problem beyond the general framework.

For  $k = 2$ , that is, for listing  $C_4$ 's, the problem is essentially understood optimally. Abboud, Khoury, Leibowitz, and Safier proved that all 4-cycles in a graph can be listed in time  $\tilde{O}(\min\{n^2, m^{4/3}\} + t)$ , and showed that this bound is conditionally tight

up to polylogarithmic factors [2]. In particular, in the sparse regime this matches the conjectural running time  $\tilde{O}(m^{4/3} + t)$  predicted by the Unbalanced Supersaturation Conjecture. Thus,  $C_4$ -listing is the only case in which the sparse output-sensitive complexity is already sharply understood.

For  $k = 3$ , the problem of listing  $C_6$ 's is more subtle. Williams and Westover [23] prove a partial form of the Unbalanced Supersaturation Conjecture for hexagons, and combine it with their layered graph framework to obtain an algorithm running in time  $\tilde{O}(m^{1.6} + t)$ . This is the current sparse state of the art for listing  $C_6$ 's, and it improves on the previous  $O(m^{5/3} + t)$  bound inherited from the general algorithmic line of Alon, Yuster, and Zwick [4, 25]. Their proof is computer-assisted, relying on an extensive case analysis in the final stage of the running-time argument [23].

At the same time, this bound is potentially not optimal. Indeed, as observed by Williams and Westover, the full Unbalanced Supersaturation Conjecture would imply a running time of  $\tilde{O}(m^{3/2} + t)$  for listing  $C_6$ 's [23]. Thus, the case  $k = 3$  already exhibits the central phenomenon of this thesis: partial progress on the unbalanced supersaturation side can lead to improved sparse listing algorithms, even when the full conjectural bound remains out of reach.



## CHAPTER 4

### A WEAK VERSION OF THE CONJECTURE

In the previous chapter, we explained the role of the Unbalanced Supersaturation Conjecture in the analysis of the even-length cycle listing algorithm. In this chapter, we prove a weaker variant of this conjecture. Although weaker than the original statement, this result is sufficient for our algorithmic application. The theorem established here will be used in Chapter 5 to derive the improved running-time bound.

#### 4.1 Statement of the Problem

**Theorem 4.1** (Weak Unbalanced Supersaturation Theorem). Let  $k \geq 2$  and let  $G = (A \cup B, E)$  be a bipartite graph with  $a := |A|$ ,  $b := |B|$ , and  $m := |E|$ . Assume without loss of generality that  $a \geq b$ . Then there exist constants  $c(k), C(k) > 0$ , depending only on  $k$ , such that if

$$m \geq C a b^{1/k},$$

then

$$\#C_{2k}(G) \geq c \frac{m^{2k}}{a^k b^k}.$$

#### 4.2 Auxiliary Lemmas

We collect the three ingredients that will be used in the proof of Theorem 4.1.

**Corollary 4.2** (Balanced bipartite case). There exist constants  $C_1(k), c_1(k) > 0$ , depending only on  $k$ , such that the following holds. Let  $H = (X \cup Y, F)$  be a bipartite graph with  $|X| = |Y| = b$ , and let  $m := |F|$ .

If

$$m \geq C_1 b^{1+1/k},$$

then

$$\#C_{2k}(H) \geq c_1 \left( \frac{m}{b} \right)^{2k}.$$

*Proof.* Let  $C_0(k), c_0(k) > 0$  be the constants from Theorem 2.9. Set

$$C_1 := C_0 2^{1+1/k} \quad \text{and} \quad c_1 := \frac{c_0}{2^{2k}}.$$

Since  $|V(H)| = 2b$ , the assumption  $m \geq C_1 b^{1+1/k}$  gives

$$m \geq C_0 (2b)^{1+1/k} = C_0 |V(H)|^{1+1/k}.$$

Therefore, by Theorem 2.9,

$$\#C_{2k}(H) \geq c_0 \left( \frac{m}{|V(H)|} \right)^{2k} = c_0 \left( \frac{m}{2b} \right)^{2k} = c_1 \left( \frac{m}{b} \right)^{2k}.$$

□

**Lemma 4.3** (Binomial ratio). If  $a \geq b \geq k \geq 1$  are integers, then

$$\frac{\binom{a}{k}}{\binom{b}{k}} \geq \left( \frac{a}{b} \right)^k.$$

*Proof.* We write

$$\frac{\binom{a}{k}}{\binom{b}{k}} = \prod_{j=0}^{k-1} \frac{a-j}{b-j}.$$

For each  $j \in \{0, 1, \dots, k-1\}$ , since  $a \geq b$  and  $b > j$ , we have

$$\frac{a-j}{b-j} - \frac{a}{b} = \frac{b(a-j) - a(b-j)}{b(b-j)} = \frac{j(a-b)}{b(b-j)} \geq 0.$$

Hence  $\frac{a-j}{b-j} \geq \frac{a}{b}$  for every  $j$ . Multiplying these  $k$  inequalities gives

$$\frac{\binom{a}{k}}{\binom{b}{k}} \geq \left( \frac{a}{b} \right)^k.$$

□

The next lemma isolates the deterministic part of the argument for a fixed partition of  $A$ .

**Lemma 4.4** (Fixed-partition lower bound). Let  $k \geq 2$ , and let  $G = (A \cup B, E)$  be a bipartite graph with  $a := |A|, b := |B|, m := |E|$ , where  $a \geq b$  and  $b \mid a$ . Set  $r := a/b$ , and partition  $A$  into  $r$  pairwise disjoint blocks

$$A = A_1 \cup \dots \cup A_r, \quad |A_i| = b \text{ for all } i \in [r].$$

For each  $i \in [r]$ , let

$$G_i := G[A_i \cup B] \quad \text{and} \quad m_i := |E(G_i)|.$$

Then there exist constants  $C(k), c(k) > 0$ , depending only on  $k$ , such that if

$$m \geq C(k) a b^{1/k},$$

then

$$\sum_{i=1}^r \#C_{2k}(G_i) \geq c(k) \frac{m^{2k}}{a^{2k-1}b}.$$

*Proof.* Let

$$L := C_1(k) b^{1+1/k},$$

where  $C_1(k)$  is the constant from Corollary 4.2. We call an index  $i \in [r]$  *heavy* if  $m_i \geq L$ , and *light* otherwise. Let  $\mathcal{H} \subseteq [r]$  denote the set of heavy indices, and set

$$m_{\mathcal{H}} := \sum_{i \in \mathcal{H}} m_i.$$

Since every light block satisfies  $m_i < L$ , it follows that

$$\sum_{i \notin \mathcal{H}} m_i < rL = \frac{a}{b} C_1(k) b^{1+1/k} = C_1(k) a b^{1/k}.$$

Consequently,

$$m_{\mathcal{H}} = m - \sum_{i \notin \mathcal{H}} m_i \geq m - C_1(k) a b^{1/k}.$$

Set  $C(k) := 2C_1(k)$ . Since

$$m \geq C(k) a b^{1/k} = 2C_1(k) a b^{1/k},$$

it follows that

$$m_{\mathcal{H}} \geq \frac{m}{2}.$$

Now let  $i \in \mathcal{H}$ . Since  $m_i \geq C_1(k) b^{1+1/k}$ , Corollary 4.2 implies that

$$\#C_{2k}(G_i) \geq c_1(k) \left( \frac{m_i}{b} \right)^{2k}.$$

Summing over the heavy indices, we obtain

$$\sum_{i=1}^r \#C_{2k}(G_i) \geq \sum_{i \in \mathcal{H}} \#C_{2k}(G_i) \geq \frac{c_1(k)}{b^{2k}} \sum_{i \in \mathcal{H}} m_i^{2k}.$$

Since the function  $x \mapsto x^{2k}$  is convex on  $\mathbb{R}_{\geq 0}$ , Jensen's inequality yields

$$\sum_{i \in \mathcal{H}} m_i^{2k} \geq |\mathcal{H}| \left( \frac{m_{\mathcal{H}}}{|\mathcal{H}|} \right)^{2k} = \frac{m_{\mathcal{H}}^{2k}}{|\mathcal{H}|^{2k-1}} \geq \frac{m_{\mathcal{H}}^{2k}}{r^{2k-1}}.$$

Therefore,

$$\sum_{i=1}^r \#C_{2k}(G_i) \geq \frac{c_1(k)}{b^{2k}} \cdot \frac{m_{\mathcal{H}}^{2k}}{r^{2k-1}} \geq \frac{c_1(k)}{b^{2k}} \cdot \frac{(m/2)^{2k}}{r^{2k-1}}.$$

Hence

$$\sum_{i=1}^r \#C_{2k}(G_i) \geq \frac{c_1(k)}{2^{2k}} \frac{m^{2k}}{r^{2k-1} b^{2k}}.$$

Finally, since  $r = a/b$ ,

$$r^{2k-1} b^{2k} = \left( \frac{a}{b} \right)^{2k-1} b^{2k} = a^{2k-1} b.$$

Set  $c(k) := c_1(k)/2^{2k}$ . It follows that

$$\sum_{i=1}^r \#C_{2k}(G_i) \geq c(k) \frac{m^{2k}}{a^{2k-1} b}.$$

□

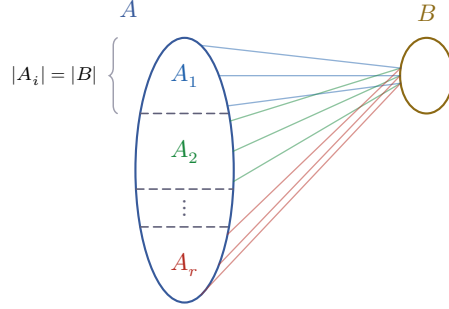


Figure 4.1: Partition of  $A$  into blocks  $A_1, \dots, A_r$  of size  $|B|$ , used to reduce the argument to balanced subinstances.

Figure 4.1 illustrates the main reduction in the proof. The set  $A$  is partitioned into  $r$  blocks  $A_1, \dots, A_r$  of size  $|B|$ , and the analysis is restricted to those blocks whose edge sets with  $B$  are above the balanced threshold and together account for at least half of the edges between  $A$  and  $B$ . This makes it possible to apply the balanced corollary blockwise while losing only a constant factor in the edge count.

### 4.3 Proof of the Main Theorem

*Proof of Theorem 4.1.* We first consider the case  $b \mid a$ . Set  $r := a/b$ , and choose uniformly at random a partition

$$A = A_1 \cup \dots \cup A_r$$

into  $r$  labeled parts of size  $b$ . Define

$$X := \sum_{i=1}^r \#C_{2k}(G[A_i \cup B]).$$

By Lemma 4.4, provided  $C(k)$  is chosen sufficiently large, every such partition satisfies

$$X \geq D(k) \frac{m^{2k}}{a^{2k-1}b}$$

for some constant  $D(k) > 0$  depending only on  $k$ . It follows that

$$\mathbb{E}[X] \geq D(k) \frac{m^{2k}}{a^{2k-1}b}.$$

Let  $\mathcal{C}$  denote the set of copies of  $C_{2k}$  in  $G$ . For each  $C \in \mathcal{C}$ , let  $\mathbf{1}_C$  be the indicator of the event that all  $k$  vertices of  $C \cap A$  lie in the same part of the partition. Then

$$X = \sum_{C \in \mathcal{C}} \mathbf{1}_C.$$

Fix  $C \in \mathcal{C}$ , and set  $S := V(C) \cap A$ , so that  $|S| = k$ . For each  $i \in [r]$ , the probability that  $S \subseteq A_i$  is

$$\frac{\binom{b}{k}}{\binom{a}{k}}.$$

Since the events  $\{S \subseteq A_i\}$  are pairwise disjoint, we obtain

$$\Pr[\mathbf{1}_C = 1] = r \frac{\binom{b}{k}}{\binom{a}{k}}.$$

Therefore

$$\mathbb{E}[X] = \sum_{C \in \mathcal{C}} \Pr[\mathbf{1}_C = 1] = r \frac{\binom{b}{k}}{\binom{a}{k}} \cdot \#C_{2k}(G).$$

Combining the two bounds on  $\mathbb{E}[X]$ , we find that

$$\#C_{2k}(G) \geq D(k) \frac{m^{2k}}{a^{2k-1}b} \cdot \frac{1}{r} \cdot \frac{\binom{a}{k}}{\binom{b}{k}}.$$

By Lemma 4.3,

$$\frac{1}{r} \cdot \frac{\binom{a}{k}}{\binom{b}{k}} \geq \frac{1}{r} \left(\frac{a}{b}\right)^k = \left(\frac{a}{b}\right)^{k-1}.$$

Hence

$$\#C_{2k}(G) \geq D(k) \frac{m^{2k}}{a^{2k-1}b} \left(\frac{a}{b}\right)^{k-1} = D(k) \frac{m^{2k}}{a^k b^k}.$$

It remains to remove the divisibility assumption. Let

$$q := \left\lfloor \frac{a}{b} \right\rfloor \quad \text{and} \quad a' := qb.$$

Then  $a' \leq a$ ,  $b \mid a'$ , and  $a' \geq a/2$ . Choose a subset  $A' \subseteq A$  with  $|A'| = a'$  maximizing the number of incident edges, and let

$$G' := G[A' \cup B], \quad m' := |E(G')|.$$

By averaging,

$$m' \geq \frac{a'}{a} m.$$

Consequently,

$$m' \geq \frac{a'}{a} C(k) a b^{1/k} = C(k) a' b^{1/k},$$

so the divisible case applies to  $G'$ . Therefore

$$\#C_{2k}(G') \geq D(k) \frac{(m')^{2k}}{(a')^k b^k} \geq D(k) \frac{(ma'/a)^{2k}}{(a')^k b^k} = D(k) \frac{m^{2k} (a')^k}{a^{2k} b^k}.$$

Since  $a' \geq a/2$ , it follows that

$$\#C_{2k}(G') \geq \frac{D(k)}{2^k} \frac{m^{2k}}{a^k b^k}.$$

Finally, every copy of  $C_{2k}$  in  $G'$  is also a copy of  $C_{2k}$  in  $G$ , and therefore

$$\#C_{2k}(G) \geq \#C_{2k}(G') \geq \frac{D(k)}{2^k} \frac{m^{2k}}{a^k b^k}.$$

Renaming the constant  $D(k)/2^k$  as  $c(k)$  completes the proof.  $\square$

For completeness, we include in Appendix A an alternative proof of the Weak Unbalanced Supersaturation Theorem.

## 4.4 Proof of the Unbalanced Supersaturation Conjecture for $k = 2$

We now give a proof using elementary graph theory for the case  $k = 2$  of the Unbalanced Supersaturation Conjecture.

**Proposition 4.5** (The case  $k = 2$ ). *Let  $G = (A \cup B, E)$  be a bipartite graph with  $a := |A|$ ,  $b := |B|$ ,  $m := |E|$ , and  $t$  copies of  $C_4$ . If*

$$m = \Omega(a + b + (ab)^{3/4}), \quad \text{then} \quad t = \Omega\left(\frac{m^4}{a^2 b^2}\right).$$

*Proof.* Assume without loss of generality that  $a \geq b$ . For each  $u \in A$ , let  $d(u)$  denote its degree. Let

$$P := \sum_{u \in A} \binom{d(u)}{2}$$

be the number of length-2 paths whose middle vertex lies in  $A$ . By convexity, and using  $m \geq 2a$ , we have

$$P \geq a \binom{m/a}{2} = \frac{m^2}{2a} - \frac{m}{2} \geq \frac{m^2}{4a}.$$

For  $x, y \in B$ , let  $p(x, y)$  be the number of common neighbours of  $x$  and  $y$  in  $A$ . Then

$$P = \sum_{\{x, y\} \in \binom{B}{2}} p(x, y).$$

Every copy of  $C_4$  is obtained by choosing two vertices  $x, y \in B$  and two common neighbours of  $x$  and  $y$  in  $A$ . Hence

$$t = \sum_{\{x, y\} \in \binom{B}{2}} \binom{p(x, y)}{2}.$$

Using convexity again, we get

$$t \geq \binom{b}{2} \binom{P/\binom{b}{2}}{2}.$$

Since  $m = \Omega(a + b + (ab)^{3/4})$ , by taking the hidden constant sufficiently large we may assume  $P \geq 2\binom{b}{2}$ . Therefore

$$\binom{P/\binom{b}{2}}{2} \geq \frac{1}{4} \left( \frac{P}{\binom{b}{2}} \right)^2.$$

Thus

$$t \geq \frac{P^2}{4\binom{b}{2}} \geq \frac{P^2}{2b^2}.$$

Using  $P \geq m^2/(4a)$ , we conclude that

$$t \geq \frac{1}{2b^2} \left( \frac{m^2}{4a} \right)^2 = \Omega\left(\frac{m^4}{a^2 b^2}\right).$$

□

# CHAPTER 5

## ALGORITHMIC CONSEQUENCES

In this chapter, we combine the weak form of the Unbalanced Supersaturation Conjecture proved in Chapter 4 with the analysis framework presented in Chapter 3. This replaces the conjectural extremal input in the Williams-Westover framework [23] by the theorem established here, yielding a refined running-time bound for even-cycle listing.

We first derive the resulting state-of-the-art consequences for sparse graphs. We then extend the discussion to hypergraphs: first through the incidence graph, which gives a global Berge-cycle listing result, and then through a labelled-shadow reduction, which gives a rooted listing result without doubling the cycle length.

### 5.1 Refined Analysis of the Existing Algorithm

In this section, we revisit the running-time analysis from Chapter 3 using the weak form of the Unbalanced Supersaturation Conjecture proved in Chapter 4. Our goal is to reproduce the analysis framework of Williams and Westover with Theorem 4.1 in place of the conjectural extremal input.

**Corollary 5.1** (Edge bound with output term). Let  $G$  be a graph, let  $A, B \subseteq V(G)$  be not necessarily disjoint, and let  $t$  denote the number of copies of  $C_{2k}$  in  $G$ . Define

$$a := \max\{|A|, |B|\}, \quad b := \min\{|A|, |B|\}, \quad \text{and} \quad m_{AB} := |E(A, B)|.$$

Then

$$m_{AB} \leq O\left(a b^{1/k} + t^{1/(2k)} \sqrt{|A||B|}\right).$$

*Proof.* We first reduce to a bipartite subgraph. Define  $C := A \cap B$ ,  $A' := A \setminus C$ , and  $B' := B \setminus C$ .

Start with left part  $L = A'$  and right part  $R = B'$ . For each vertex  $c \in C$ , independently place  $c$  in  $L$  with probability  $1/2$  and in  $R$  otherwise. Let  $H$  be the subgraph of  $G$  consisting only of the edges with one endpoint in  $L$  and the other in  $R$ . Then  $H$  is bipartite with parts  $L$  and  $R$ .

Each edge counted by  $m_{AB}$  is retained in  $H$  with probability at least  $1/2$ . Therefore,

$$\mathbb{E}|E(H)| \geq \frac{m_{AB}}{2}.$$

It follows that there exists a choice of the random partition for which

$$|E(H)| \geq \frac{m_{AB}}{2}.$$

Fix such a choice. Since  $L \subseteq A$  and  $R \subseteq B$ , we have  $|L| \leq |A|$  and  $|R| \leq |B|$ .

Let  $t_H$  be the number of copies of  $C_{2k}$  in  $H$ . Because  $H \subseteq G$ , every copy of  $C_{2k}$  in  $H$  is also a copy of  $C_{2k}$  in  $G$ , and hence  $t_H \leq t$ .

Now set

$$a_H := \max\{|L|, |R|\}, \quad b_H := \min\{|L|, |R|\}.$$

Then  $a_H \leq a$  and  $b_H \leq b$ .

We distinguish two cases.

**Case 1:** If  $|E(H)| \leq C(k) a_H b_H^{1/k}$ , then

$$m_{AB} \leq 2|E(H)| \leq 2C(k) a_H b_H^{1/k} \leq O(a b^{1/k}).$$

**Case 2:** Assume now that  $|E(H)| > C(k) a_H b_H^{1/k}$ . Since  $H$  is bipartite with parts  $L$  and  $R$ , Theorem 4.1 applies to  $H$ . Therefore,

$$t_H \geq c(k) \frac{|E(H)|^{2k}}{a_H^k b_H^k}.$$

Using  $t \geq t_H$ ,  $|E(H)| \geq m_{AB}/2$ ,  $a_H \leq a$ , and  $b_H \leq b$ , we obtain

$$t \geq c(k) \frac{(m_{AB}/2)^{2k}}{a^k b^k}.$$

Rearranging this inequality gives

$$m_{AB} \leq O\left(t^{1/(2k)} \sqrt{ab}\right) = O\left(t^{1/(2k)} \sqrt{|A||B|}\right).$$

Combining the two cases, we conclude that

$$m_{AB} \leq O\left(a b^{1/k} + t^{1/(2k)} \sqrt{|A||B|}\right).$$

□

**Lemma 5.2** (Layer-to-layer degree ratio bound). Let  $G$  be a graph with  $\Delta(G) \leq m^{\frac{k+1}{k^2+1}}$ , let  $A \subseteq V(G)$ , and let  $d \leq \Delta$ . Define

$$B := \{v \in V(G) : |N_A(v)| \in [d/2, d]\}.$$

Then

$$d \sqrt{\frac{|B|}{|A|}} \leq O\left(m^{\frac{k}{k^2+1}} + t^{1/(2k)}\right).$$

*Proof.* Define

$$Q := d \sqrt{\frac{|B|}{|A|}}.$$

By the definition of  $B$ , every vertex  $v \in B$  satisfies  $|N_A(v)| \in [d/2, d]$ . Hence

$$\frac{d}{2}|B| \leq m_{AB} \leq d|B|.$$

Dividing the lower bound by  $\sqrt{|A||B|}$ , we obtain

$$\frac{Q}{2} = \frac{d}{2} \sqrt{\frac{|B|}{|A|}} \leq \frac{m_{AB}}{\sqrt{|A||B|}},$$

and therefore

$$Q \leq 2 \frac{m_{AB}}{\sqrt{|A||B|}}.$$

By Corollary 5.1,

$$Q \leq O\left(\frac{a b^{1/k}}{\sqrt{|A||B|}} + t^{1/(2k)}\right),$$

where  $a = \max\{|A|, |B|\}$  and  $b = \min\{|A|, |B|\}$ .

We now distinguish two cases.

**Case 1:**  $|B| \leq |A|$ . Then  $a = |A|$  and  $b = |B|$ , so

$$\frac{a b^{1/k}}{\sqrt{|A||B|}} = \sqrt{|A|} |B|^{1/k-1/2}.$$

Using  $Q = d\sqrt{|B|/|A|}$ , we rewrite this as

$$\sqrt{|A|} |B|^{1/k-1/2} = \frac{d |B|^{1/k}}{Q}.$$

Thus

$$Q \leq O\left(\frac{d |B|^{1/k}}{Q} + t^{1/(2k)}\right),$$

and hence

$$Q^2 \leq O(d |B|^{1/k}) + O(t^{1/(2k)} Q).$$

Solving this quadratic inequality gives

$$Q \leq O\left(\sqrt{d |B|^{1/k}} + t^{1/(2k)}\right).$$

Since  $m_{AB} \leq m$  and  $(d/2)|B| \leq m_{AB}$ , we have  $|B| \leq 2m/d$ . Therefore

$$\sqrt{d |B|^{1/k}} \leq O((md^{k-1})^{1/(2k)}) \leq O((m\Delta^{k-1})^{1/(2k)}) = O\left(m^{\frac{k}{k^2+1}}\right).$$

Thus

$$Q \leq O\left(m^{\frac{k}{k^2+1}} + t^{1/(2k)}\right).$$

**Case 2:**  $|B| \geq |A|$ . Set

$$X := d^2 \frac{|B|}{|A|}, \quad \text{so that} \quad Q = \sqrt{X}.$$

From  $d|B| \leq 2m_{AB}$ , we get

$$X = d^2 \frac{|B|}{|A|} \leq 4 \frac{m_{AB}^2}{|A||B|}.$$

Since now  $a = |B|$  and  $b = |A|$ , Corollary 5.1 yields

$$m_{AB} \leq O\left(|B||A|^{1/k} + t^{1/(2k)}\sqrt{|A||B|}\right).$$

Dividing by  $\sqrt{|A||B|}$  and squaring, we obtain

$$\frac{m_{AB}^2}{|A||B|} \leq O(|B||A|^{2/k-1} + t^{1/k}),$$

and hence

$$X \leq O(|B||A|^{2/k-1} + t^{1/k}).$$

Rewrite the mixed term as

$$|B||A|^{2/k-1} = |B|^{2/k} \left(\frac{|B|}{|A|}\right)^{1-2/k}.$$

From  $m_{AB} \leq m$  and  $(d/2)|B| \leq m_{AB}$ , we get

$$|B| \leq \frac{2m}{d}.$$

Also, since  $m_{AB} \leq \Delta|A|$ , we have

$$\frac{|B|}{|A|} \leq \frac{2\Delta}{d}.$$

Substituting these bounds gives

$$|B||A|^{2/k-1} \leq O(m^{2/k} \Delta^{1-2/k} d^{-1}).$$

Moreover,

$$X = d^2 \frac{|B|}{|A|} \leq 2d\Delta.$$

Combining the two bounds, we get

$$X \leq O(t^{1/k} + \min\{2d\Delta, m^{2/k} \Delta^{1-2/k} d^{-1}\}).$$

Using  $\min\{u, v\} \leq \sqrt{uv}$ , it follows that

$$X \leq O(t^{1/k} + m^{1/k} \Delta^{1-1/k}).$$

Since

$$m^{1/k} \Delta^{1-1/k} = m^{1/k} \left(m^{\frac{k+1}{k^2+1}}\right)^{1-1/k} = m^{\frac{2k}{k^2+1}},$$

we conclude that

$$X \leq O\left(t^{1/k} + m^{\frac{2k}{k^2+1}}\right).$$

Taking square roots gives

$$Q \leq O\left(m^{\frac{k}{k^2+1}} + t^{1/(2k)}\right).$$

This completes the proof. □

**Theorem 5.3** (Bound on capped  $k$ -walks). Let  $G$  be a graph with  $\Delta(G) \leq m^{\frac{k+1}{k^2+1}}$ . Let  $W$  denote the number of capped  $k$ -walks in  $G$ , then

$$W \leq \tilde{O}\left(m^{\frac{2k^2-k+1}{k^2+1}} + t\right).$$

*Proof.* Set

$$M := m^{\frac{k}{k^2+1}} + t^{1/(2k)}.$$

Apply the layered decomposition lemma 3.5 from Chapter 3 to obtain  $V^*$ ,  $d^*$ ,  $G'$ ,  $X_1, \dots, X_k \subseteq V(G')$ , and  $d_1, \dots, d_k$  with the stated properties. The number of  $k$ -walks in

$$X_1 \times \dots \times X_k \times V(G')$$

is at most

$$|X_1| \prod_{j=1}^k d_j.$$

For each  $i \in [k-1]$ , we apply Lemma 5.2 with  $A = X_{i+1}$ ,  $B = X_i$ , and  $d = d_i$ . Since every vertex of  $X_i$  has between  $d_i/2$  and  $d_i$  neighbors in  $X_{i+1}$ , Lemma 5.2 gives

$$d_i \sqrt{|X_i|} \leq O\left(M \sqrt{|X_{i+1}|}\right).$$

Repeatedly applying this bound for  $i = 1, \dots, k-1$  yields

$$|X_1| \prod_{j=1}^k d_j \leq \sqrt{|X_1| |X_k|} d_k \cdot O(M^{k-1}).$$

We now bound the factor  $\sqrt{|X_1| |X_k|} d_k$ . First, since every vertex of  $X_k$  has degree at least  $d_k/2$  in  $G'$ , we have

$$|X_k| d_k \leq O(m).$$

Second,  $X_1 \subseteq V^*$  and  $d^* \geq d_k$ , while every vertex of  $V^*$  has degree at least  $d^*/2$ . Hence

$$|X_1| d_k \leq |X_1| d^* \leq O(m).$$

Therefore

$$\sqrt{|X_1| |X_k|} d_k \leq \sqrt{(|X_1| d_k)(|X_k| d_k)} \leq O(m).$$

Expanding  $M$  and applying Hölder's inequality yields

$$m M^{k-1} = m \left( m^{\frac{k}{k^2+1}} + t^{1/(2k)} \right)^{k-1} \leq O\left( m^{\frac{2k^2-k+1}{k^2+1}} + t \right).$$

The mixed term is controlled by Young's inequality, since  $m t^{\frac{k-1}{2k}} \leq O\left( m^{\frac{2k}{k+1}} + t \right)$ . Therefore,

$$|X_1| \prod_{j=1}^k d_j \leq O\left( m^{\frac{2k^2-k+1}{k^2+1}} + t \right).$$

Now Lemma 3.5 implies that the number  $W$  of capped  $k$ -walks in  $G$  is at most  $\tilde{O}(1)$  times the number of  $k$ -walks in

$$X_1 \times \dots \times X_k \times V(G'),$$

and hence

$$W \leq \tilde{O}\left(|X_1| \prod_{j=1}^k d_j\right) \leq \tilde{O}\left(m^{\frac{2k^2-k+1}{k^2+1}} + t\right).$$

□

**Corollary 5.4** (Final running time bound). The running time of the even-cycle listing algorithm satisfies

$$T \leq \tilde{O}\left(m^{\frac{2k^2-k+1}{k^2+1}} + t\right).$$

Equivalently,

$$T \leq \tilde{O}\left(m^{2-\frac{1}{k}-\frac{k-1}{k(k^2+1)}} + t\right).$$

*Proof.* By the algorithm's running time decomposition from Chapter 3,

$$T \leq \tilde{O}\left(\frac{m^2}{\Delta} + W + t\right).$$

Set  $\Delta := m^{\frac{k+1}{k^2+1}}$ , thus

$$\frac{m^2}{\Delta} = m^{2-\frac{k+1}{k^2+1}} = m^{\frac{2k^2-k+1}{k^2+1}}.$$

Combining this with Theorem 5.3 gives

$$T \leq \tilde{O}\left(m^{\frac{2k^2-k+1}{k^2+1}} + t\right).$$

which gives the second form of the bound. □

Additionally, combining Proposition 4.5 with Corollary 29 of [23], we obtain the following consequence.

**Proposition 5.5.** *Given an  $n$ -vertex,  $m$ -edge graph  $G$ , all copies of  $C_4$  in  $G$  can be listed in*

$$\tilde{O}(\min\{m^{4/3}, n^2\} + t)$$

*time, where  $t$  is the number of copies of  $C_4$  in  $G$ . Moreover, this running time is conditionally optimal up to polylogarithmic factors under standard fine-grained assumptions.*

*Proof.* Proposition 4.5 proves the  $k = 2$  case of the Unbalanced Supersaturation Conjecture. Therefore, Corollary 29 of Williams and Westover [23] applies to the listing algorithm of Theorem 3.4. This gives an algorithm for listing all copies of  $C_4$  in time

$$\tilde{O}(\min\{m^{4/3}, n^2\} + t).$$

The conditional optimality of this bound, up to polylogarithmic factors, follows from the known fine-grained lower bounds for  $C_4$ -listing [2]. □

## 5.2 State-of-the-Art Consequences

We now summarize what the preceding analysis gives for the current state of the art in even-cycle listing.

For  $k = 2$ , that is, for listing 4-cycles, our proof gives a more compact derivation of the conditionally optimal bound

$$\tilde{O}(\min\{m^{4/3}, n^2\} + t).$$

This places the  $C_4$ -listing result inside the same general framework that we use for longer even cycles.

For  $k = 3$ , that is, for listing 6-cycles, Corollary 5.4 gives

$$\tilde{O}(m^{1.6} + t).$$

This matches the current best sparse-graph bound of Williams and Westover [23], while avoiding the computer-aided part of their analysis.

For every  $k \geq 4$ , Corollary 5.4 improves the previously known sparse bound

$$O(m^{2-1/k} + t),$$

which follows from the work of Alon, Yuster, and Zwick [4] as noted by Williams and Westover [23]. Thus, the weak form of the Unbalanced Supersaturation Conjecture proved in Chapter 4 is already strong enough to improve the state of the art for listing  $C_{2k}$ 's for every  $k \geq 4$ .

## 5.3 Extension to Hypergraphs

We now describe two extensions of the graph result to hypergraphs. The first one is global and follows immediately from the incidence graph. The second one is more direct and uses the labelled 2-shadow of the hypergraph. This second reduction avoids the doubling of the cycle length, but it applies to the rooted version of the problem, where the input also contains a prescribed vertex  $v$ .

Throughout this section, let  $\mathcal{H} = (V, \mathcal{E})$  be a hypergraph. We write  $\|\mathcal{H}\| := \sum_{e \in \mathcal{E}} |e|$  for the size of  $\mathcal{H}$ , and  $r := \max_{e \in \mathcal{E}} |e|$  for its rank.

### 5.3.1 Global Listing via the Incidence Graph

The incidence graph  $I(\mathcal{H})$  is the bipartite graph with vertex set  $V \cup \mathcal{E}$ , where  $v \in V$  is adjacent to  $e \in \mathcal{E}$  if and only if  $v \in e$ . The key observation is that Berge cycles in  $\mathcal{H}$  are exactly ordinary cycles in this incidence graph.

**Lemma 5.6.** Let  $\mathcal{H} = (V, \mathcal{E})$  be a hypergraph. For every  $\ell \geq 2$ , Berge cycles of length  $\ell$  in  $\mathcal{H}$  are in one-to-one correspondence with cycles of length  $2\ell$  in  $I(\mathcal{H})$ .

*Proof.* A Berge cycle

$$v_1, e_1, v_2, e_2, \dots, v_\ell, e_\ell$$

corresponds to the cycle

$$v_1, e_1, v_2, e_2, \dots, v_\ell, e_\ell, v_1$$

in the incidence graph. Conversely, every cycle of length  $2\ell$  in  $I(\mathcal{H})$  alternates between vertices of  $V$  and hyperedges of  $\mathcal{E}$ , and therefore gives a Berge cycle of length  $\ell$  in  $\mathcal{H}$ . The distinctness conditions are the same in both representations.  $\square$

**Corollary 5.7.** Let  $\mathcal{H} = (V, \mathcal{E})$  be a hypergraph, let

$$m := |\mathcal{E}|, \quad M := |E(I(\mathcal{H}))| = \sum_{e \in \mathcal{E}} |e| = \|\mathcal{H}\|,$$

and let  $t$  denote the number of Berge cycles of length  $\ell$  in  $\mathcal{H}$ . Then all Berge cycles of length  $\ell$  in  $\mathcal{H}$  can be listed in time

$$\tilde{O}\left(M^{2 - \frac{\ell+1}{\ell^2+1}} + t\right).$$

In particular, if  $\mathcal{H}$  has rank at most  $r$ , then  $M \leq rm$ , and the running time becomes

$$\tilde{O}\left((rm)^{2 - \frac{\ell+1}{\ell^2+1}} + t\right).$$

*Proof.* By Lemma 5.6, Berge cycles of length  $\ell$  in  $\mathcal{H}$  are in one-to-one correspondence with cycles of length  $2\ell$  in the incidence graph  $I(\mathcal{H})$ . Applying Corollary 5.4 to  $I(\mathcal{H})$ , with  $k = \ell$ , gives

$$\tilde{O}\left(M^{2 - \frac{\ell+1}{\ell^2+1}} + t\right),$$

because  $I(\mathcal{H})$  has  $M$  edges.

If  $\mathcal{H}$  has rank at most  $r$ , then

$$M = \sum_{e \in \mathcal{E}} |e| \leq r|\mathcal{E}| = rm.$$

Substituting this into the previous bound gives the claimed rank- $r$  bound.  $\square$

This reduction is completely general, but it has one drawback: a Berge cycle of length  $\ell$  becomes a graph cycle of length  $2\ell$ . Thus, for even-length Berge cycles, the exponent is governed by the doubled length. We next describe a more direct reduction for rooted Berge-cycle listing.

### 5.3.2 Rooted Listing via the Labelled 2-Shadow

We now fix a vertex  $v \in V$  and consider the problem of listing all Berge cycles of length  $\ell$  that contain  $v$ . For this rooted problem, we can work directly with the labelled 2-shadow of the hypergraph.

The labelled 2-shadow of  $\mathcal{H}$ , denoted by  $S_2(\mathcal{H})$ , is the multigraph defined as follows. Its vertex set is  $V$ . For every hyperedge  $e \in \mathcal{E}$  and every unordered pair  $\{x, y\} \subseteq e$ , we add one multiedge between  $x$  and  $y$ , labelled by  $e$ . Thus a multiedge of  $S_2(\mathcal{H})$  is formally a triple

$$\eta = (x, y, e), \quad x, y \in e, \quad x \neq y,$$

and its label is

$$\lambda(\eta) = e.$$

The number of multiedges in  $S_2(\mathcal{H})$  is

$$N_2(\mathcal{H}) := \sum_{e \in \mathcal{E}} \binom{|e|}{2}.$$

If  $\mathcal{H}$  has rank at most  $r$ , then

$$N_2(\mathcal{H}) = \sum_{e \in \mathcal{E}} \binom{|e|}{2} \leq \frac{r-1}{2} \sum_{e \in \mathcal{E}} |e| = O(r\|\mathcal{H}\|).$$

**Lemma 5.8.** Let  $\mathcal{H} = (V, \mathcal{E})$  be a hypergraph, and let  $v \in V$ . Berge cycles of length  $\ell$  in  $\mathcal{H}$  that contain  $v$  are represented by simple cycles of length  $\ell$  in  $S_2(\mathcal{H})$  that contain  $v$  and whose edge labels are pairwise distinct, and conversely.

*Proof.* A Berge cycle

$$v = x_0, e_1, x_1, e_2, \dots, e_\ell, x_\ell = x_0$$

uses distinct vertices  $x_0, \dots, x_{\ell-1}$  and distinct hyperedges  $e_1, \dots, e_\ell$ , with  $\{x_{i-1}, x_i\} \subseteq e_i$ . Hence the labelled 2-shadow contains the multiedges  $(x_{i-1}, x_i, e_i)$ , which form a simple cycle through  $v$  with pairwise distinct labels.

Conversely, a simple cycle in the labelled 2-shadow with pairwise distinct labels directly specifies distinct vertices and distinct hyperedges satisfying the definition of a Berge cycle.  $\square$

It remains to list simple cycles through  $v$  in the labelled shadow while enforcing that the edge labels are distinct. Since  $\ell$  is fixed, we do this by combining color-coding with constant-delay enumeration of a fixed free-connex acyclic conjunctive query with disequalities.

Fix a coloring

$$\chi : V \setminus \{v\} \rightarrow [\ell - 1].$$

A cycle through  $v$  is called colorful if its  $\ell - 1$  vertices different from  $v$  receive pairwise distinct colors. For each permutation

$$\pi = (\pi(1), \dots, \pi(\ell - 1))$$

of  $[\ell - 1]$ , construct a layered multigraph  $G_{\chi, \pi}$  with layers

$$L_0, L_1, \dots, L_\ell$$

as follows:

$$L_0 = \{v\}, \quad L_\ell = \{v\},$$

and, for  $1 \leq i \leq \ell - 1$ ,

$$L_i = \{x \in V \setminus \{v\} : \chi(x) = \pi(i)\}.$$

We keep only the multiedges of  $S_2(\mathcal{H})$  that go between consecutive layers. Therefore every layer-respecting path from  $L_0$  to  $L_\ell$  has the form

$$v = x_0, x_1, \dots, x_{\ell-1}, x_\ell = v.$$

For a fixed coloring  $\chi$  and a fixed permutation  $\pi$ , the colorful Berge cycles through  $v$  whose colors appear in the order  $\pi$  are exactly the  $v$ -to- $v$  paths in  $G_{\chi, \pi}$  whose edge labels are pairwise distinct.

To enumerate these paths, define one relation for each layer transition. For each  $i \in [\ell]$ , let

$$R_i(x_{i-1}, x_i, \lambda_i, \eta_i)$$

contain one tuple for every multiedge  $\eta_i$  from  $L_{i-1}$  to  $L_i$ , where

$$L_0 = L_\ell = \{v\}.$$

The variables  $x_{i-1}$  and  $x_i$  store the endpoints of the multiedge,  $\lambda_i$  stores its hyperedge label, and  $\eta_i$  stores the multiedge itself. Since  $L_0 = L_\ell = \{v\}$ , the endpoint variables  $x_0$  and  $x_\ell$  are fixed to  $v$ .

Now consider the query

$$Q_{\chi, \pi} := \bigwedge_{i=1}^{\ell} R_i(x_{i-1}, x_i, \lambda_i, \eta_i) \wedge \bigwedge_{1 \leq i < j \leq \ell} (\lambda_i \neq \lambda_j), \quad x_0 = x_\ell = v.$$

After substituting  $x_0 = x_\ell = v$ , the positive part of  $Q_{\chi, \pi}$  is path-shaped, with join tree

$$R_1 - R_2 - \dots - R_\ell.$$

The free variables specify the whole labelled path, so they are covered by a connected part of this join tree. Therefore  $Q_{\chi, \pi}$  is a fixed free-connex acyclic conjunctive query with disequalities, that is, a fixed query in the class  $CCQ^\neq$  of Bagan, Durand, and Grandjean [5].

By their constant-delay enumeration theorem for  $CCQ^\neq$ , after linear preprocessing in the size of the relations, all answers to  $Q_{\chi, \pi}$  can be enumerated with constant delay. Since  $\ell$  is fixed, the hidden constants depend only on  $\ell$ .

**Theorem 5.9.** Fix  $\ell \geq 3$ . Let  $\mathcal{H} = (V, \mathcal{E})$  be a hypergraph, let  $v \in V$ , and let  $t_v$  denote the number of Berge cycles of length  $\ell$  in  $\mathcal{H}$  that contain  $v$ . Then all such cycles can be listed in time

$$\tilde{O}_\ell \left( \sum_{e \in \mathcal{E}} \binom{|e|}{2} + t_v \right).$$

In particular, if  $\mathcal{H}$  has rank at most  $r$ , then the running time is

$$\tilde{O}_\ell(r \|\mathcal{H}\| + t_v).$$

*Proof.* First construct the labelled 2-shadow  $S_2(\mathcal{H})$ . This takes time

$$O \left( \sum_{e \in \mathcal{E}} \binom{|e|}{2} \right).$$

We use a standard derandomized color-coding family from  $V \setminus \{v\}$  to  $[\ell - 1]$ . Since  $\ell$  is fixed, this contributes only a polylogarithmic factor in the input size. The family guarantees that, for every set of  $\ell - 1$  vertices, some coloring makes these vertices colorful.

For each coloring  $\chi$  in this family and each permutation  $\pi$  of  $[\ell - 1]$ , we construct the layered multigraph  $G_{\chi, \pi}$  and the relations  $R_1, \dots, R_\ell$ . Their total size is at most

$$\sum_{e \in \mathcal{E}} \binom{|e|}{2}.$$

As explained above, the query  $Q_{\chi, \pi}$  is a fixed free-connex acyclic conjunctive query with disequalities. Hence, by the constant-delay enumeration theorem of Bagan, Durand, and Grandjean [5], its answers can be enumerated after linear preprocessing with constant delay.

Each answer gives a  $v$ -to- $v$  path in the layered shadow whose edge labels are pairwise distinct. Therefore, by Lemma 5.8, each answer gives a Berge cycle of length  $\ell$

through  $v$ . Conversely, every Berge cycle of length  $\ell$  through  $v$  becomes colorful for at least one coloring in the family and appears in the layered graph corresponding to the permutation matching its color order.

The same Berge cycle may be produced for more than one coloring and possibly in both orientations. We remove duplicates by storing a canonical representation of each reported Berge cycle. Since  $\ell$  is fixed, computing and checking this representation costs  $\tilde{O}_\ell(1)$  per produced answer.

Thus the total running time is  $\tilde{O}_\ell\left(\sum_{e \in \mathcal{E}} \binom{|e|}{2} + t_v\right)$ .

If the rank is at most  $r$ , then  $\sum_{e \in \mathcal{E}} \binom{|e|}{2} \leq O(r\|\mathcal{H}\|)$ , which gives the claimed rank- $r$  bound.  $\square$

The two reductions therefore give complementary hypergraph extensions. The incidence graph gives a global listing result for all Berge cycles of length  $\ell$ , but it turns them into graph cycles of length  $2\ell$ . The labelled-shadow reduction avoids this length doubling, but in the form above it solves the rooted problem, listing only the Berge cycles of length  $\ell$  that contain a prescribed vertex  $v$ .



## CHAPTER 6

## CONCLUSION AND OPEN PROBLEMS

### 6.1 Summary of Contributions

In this thesis, we studied the problem of listing even cycles in sparse graphs and its connection with extremal graph theory.

Our first contribution was extremal. We proved a weak form of the Unbalanced Supersaturation Conjecture for even cycles in bipartite graphs, obtaining a quantitative lower bound on the number of copies of  $C_{2k}$  in terms of the number of edges and the sizes of the two sides of the bipartition. We also gave an elementary graph-theoretic proof of the folklore case  $k = 2$ , thereby proving the conjecture for  $C_4$ 's.

Our second contribution was algorithmic. By combining the weak supersaturation result with the analysis framework of Williams and Westover, we obtained a refined running-time analysis for the existing even-cycle listing algorithm. For  $k = 2$ , the conditionally optimal algorithm for listing  $C_4$ 's follows directly from the  $k = 2$  solution of the Unbalanced Supersaturation Conjecture, placing the known  $C_4$ -listing bound inside the same unified framework used for longer even cycles. This recovers the conditionally optimal  $\tilde{O}(\min\{m^{4/3}, n^2\} + t)$  bound for listing  $C_4$ 's. For  $k = 3$ , our analysis matches the current best sparse-graph bound  $\tilde{O}(m^{1.6} + t)$  for listing  $C_6$ 's, while avoiding the computer-aided part of the previous analysis. For every  $k \geq 4$ , it improves the previously known sparse bound for listing  $C_{2k}$ 's.

Our third contribution was to extend this viewpoint to hypergraphs. First, using the correspondence between Berge cycles of length  $\ell$  in a hypergraph and cycles of length  $2\ell$  in its incidence graph, we obtained a global listing result for Berge cycles of arbitrary fixed length. Second, for the rooted version of the problem, we gave a more direct reduction through the labelled 2-shadow of the hypergraph. This avoids the doubling of the cycle length and lists all Berge cycles of length  $\ell$  that contain a prescribed vertex.

### 6.2 Open Problems

The results of this thesis suggest several natural directions for future work.

A first direction is to bring different tools into the extremal side of the problem. The proof in this thesis is combinatorial. It would be interesting to understand whether probabilistic or algebraic methods can give alternative proofs of the relevant extremal phenomena. A natural starting point would be to reprove the existential threshold of Dahlgaard, Knudsen, and Stöckel, Theorem 2.10, using such methods. A more ambitious goal would be to apply these tools directly to unbalanced supersaturation.

A second direction is to strengthen the hypergraph generalization. The labelled 2-shadow reduction developed in this thesis avoids the length doubling of the incidence graph, but only for the rooted problem: it lists Berge cycles of length  $\ell$  that contain a prescribed vertex  $v$ . The natural next step is to turn this rooted algorithm into a global listing algorithm, in the same spirit as the reduction of Dahlgaard, Knudsen, and Stöckel from finding cycles through a fixed vertex to finding cycles in the whole graph. In the hypergraph setting, this would mean using the labelled-shadow routine as the local subproblem and then designing an ordering or capped-walk framework that lists all Berge cycles without passing through the incidence graph. Such a result would give a more faithful hypergraph analogue of the graph algorithm, since a Berge cycle of length  $2k$  would be handled at length  $2k$ , rather than being converted into a graph cycle of length  $4k$ .

Finally, the most important open problem remains the full Unbalanced Supersaturation Conjecture. In this thesis, we proved a weaker form that is already sufficient to improve the analysis of even-cycle listing algorithms. Proving the conjecture in its original form would complete the extremal input required by the Williams-Westover framework and would yield the conjectured sparse running time for listing fixed even cycles.

### 6.3 Remarks on Subsequent Work

The results of this thesis form the basis of a subsequent manuscript, titled *Listing Even Cycles Beyond the Submodular-Width Barrier*, which is currently under submission. That work develops the algorithmic consequences further in three directions.

First, it removes the polylogarithmic factors appearing in the framework of this thesis by generalizing the capped-walk framework of Dahlgaard, Knudsen, and Stöckel [8] from detection to listing. The resulting bounds therefore match the corresponding sparse cycle-listing exponents without the logarithmic losses introduced by the layered decomposition used here.

Second, it gives a different proof that the full Unbalanced Supersaturation Conjecture implies the conditionally optimal running time  $O(m^{2k/(k+1)} + t)$  for listing  $C_{2k}$ 's, again without the polylogarithmic losses coming from the layered decomposition used by Williams and Westover [23]. In this sense, the conjecture captures exactly the extremal input needed to recover the optimal cycle-listing exponent.

Third, the manuscript reformulates the result in the language of database theory. In this formulation, listing even cycles is viewed as evaluating cycle conjunctive queries. This gives a database-theoretic interpretation of the algorithmic result: for these cyclic queries, the analysis goes beyond the general bound obtained from the submodular-width framework. Thus, the follow-up work can be viewed both as a fine-grained graph-algorithmic improvement and as an output-sensitive enumeration result for cyclic conjunctive queries.

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## APPENDIX A

### ALTERNATIVE PROOF OF WEAK UNBALANCED SUPERSATURATION THEOREM

We give an alternative proof of Theorem 4.1, based on summing over all balanced partitions of the larger vertex class and then dividing by the multiplicity with which a fixed copy of  $C_{2k}$  is counted. As shown in the main proof, it is enough to treat the case  $b \mid a$ ; the general case is reduced to this one.

*Alternative proof of Theorem 4.1.* We first consider the case  $b \mid a$ , and write

$$r := \frac{a}{b}.$$

Let  $\mathcal{P}$  denote the family of all ordered partitions

$$(A_1, \dots, A_r)$$

of  $A$  into  $r$  pairwise disjoint parts of size  $b$ .

For each partition  $\mathbf{A} = (A_1, \dots, A_r) \in \mathcal{P}$ , define

$$X(\mathbf{A}) := \sum_{i=1}^r \#C_{2k}(G[A_i \cup B]).$$

By Lemma 4.4, there exist constants  $C(k), d(k) > 0$ , depending only on  $k$ , such that if

$$m \geq C(k) a b^{1/k},$$

then every partition  $\mathbf{A} \in \mathcal{P}$  satisfies

$$X(\mathbf{A}) \geq d(k) \frac{m^{2k}}{a^{2k-1}b}. \quad (\text{A.1})$$

Summing (A.1) over all partitions in  $\mathcal{P}$ , we obtain

$$A := \sum_{\mathbf{A} \in \mathcal{P}} X(\mathbf{A}) \geq |\mathcal{P}| d(k) \frac{m^{2k}}{a^{2k-1}b}. \quad (\text{A.2})$$

We now calculate how many times a fixed copy of  $C_{2k}$  is counted in  $A$ . Let  $C$  be a fixed copy of  $C_{2k}$  in  $G$ . Since  $G$  is bipartite, the cycle  $C$  contains exactly  $k$  vertices from  $A$ . Moreover,  $C$  contributes to  $X(\mathbf{A})$  if and only if all vertices of  $V(C) \cap A$  lie in the same part of the partition  $\mathbf{A} = (A_1, \dots, A_r)$ .

Let  $M$  denote the number of ordered partitions in  $\mathcal{P}$  for which this happens. The total number of ordered partitions of  $A$  into  $r$  parts of size  $b$  is

$$|\mathcal{P}| = \binom{a}{b, \dots, b} = \frac{a!}{(b!)^r}.$$

On the other hand, to count the partitions for which  $V(C) \cap A$  is contained in a single part, we first choose the index  $i \in [r]$  of that part, and then distribute the remaining  $a - k$  vertices so that part  $A_i$  receives  $b - k$  of them and each of the other  $r - 1$  parts receives  $b$ . Thus

$$M = r \binom{a - k}{b - k, b, \dots, b} = r \frac{(a - k)!}{(b - k)! (b!)^{r-1}}.$$

Therefore

$$\frac{M}{|\mathcal{P}|} = r \cdot \frac{(a - k)!}{(b - k)! (b!)^{r-1}} \cdot \frac{(b!)^r}{a!} = r \cdot \frac{b!}{(b - k)!} \cdot \frac{(a - k)!}{a!} = r \frac{\binom{b}{k}}{\binom{a}{k}}.$$

By Lemma 4.3,

$$\frac{\binom{a}{k}}{\binom{b}{k}} \geq \left(\frac{a}{b}\right)^k = r^k,$$

and hence

$$\frac{M}{|\mathcal{P}|} \leq \frac{1}{r^{k-1}}.$$

Equivalently,

$$M \leq \frac{|\mathcal{P}|}{r^{k-1}}.$$

Since every copy of  $C_{2k}$  in  $G$  is counted at most  $M$  times in the sum  $A$ , it follows from (A.2) that

$$\#C_{2k}(G) \geq \frac{A}{M} \geq \frac{|\mathcal{P}|}{M} d(k) \frac{m^{2k}}{a^{2k-1}b}.$$

Using the bound on  $M$ , we obtain

$$\#C_{2k}(G) \geq r^{k-1} d(k) \frac{m^{2k}}{a^{2k-1}b}.$$

Since  $a = rb$ , this becomes

$$\#C_{2k}(G) \geq d(k) \frac{m^{2k}}{a^k b^k}.$$

It remains to remove the divisibility assumption. Let

$$q := \left\lfloor \frac{a}{b} \right\rfloor \quad \text{and} \quad a' := qb.$$

Then  $a' \leq a$ ,  $b \mid a'$ , and  $a' \geq a/2$ . Choose a subset  $A' \subseteq A$  with  $|A'| = a'$  maximizing the number of incident edges, and let

$$G' := G[A' \cup B], \quad m' := |E(G')|.$$

By averaging,

$$m' \geq \frac{a'}{a} m.$$

Consequently,

$$m' \geq \frac{a'}{a} C(k) a b^{1/k} = C(k) a' b^{1/k},$$

so the divisible case applies to  $G'$ . Therefore

$$\#C_{2k}(G') \geq d(k) \frac{(m')^{2k}}{(a')^k b^k} \geq d(k) \frac{(ma'/a)^{2k}}{(a')^k b^k} = d(k) \frac{m^{2k} (a')^k}{a^{2k} b^k}.$$

Since  $a' \geq a/2$ , it follows that

$$\#C_{2k}(G') \geq \frac{d(k)}{2^k} \frac{m^{2k}}{a^k b^k}.$$

Finally, every copy of  $C_{2k}$  in  $G'$  is also a copy of  $C_{2k}$  in  $G$ , and therefore

$$\#C_{2k}(G) \geq \#C_{2k}(G') \geq \frac{d(k)}{2^k} \frac{m^{2k}}{a^k b^k}.$$

Renaming the constant  $d(k)/2^k$  as  $c(k)$  completes the proof.  $\square$